Purwadhika Digital Technology School | Digital For Executive

Data Analysis

Session 14-15: Tableau

Digital for Executive

What is Tableau

(Tableau Public)

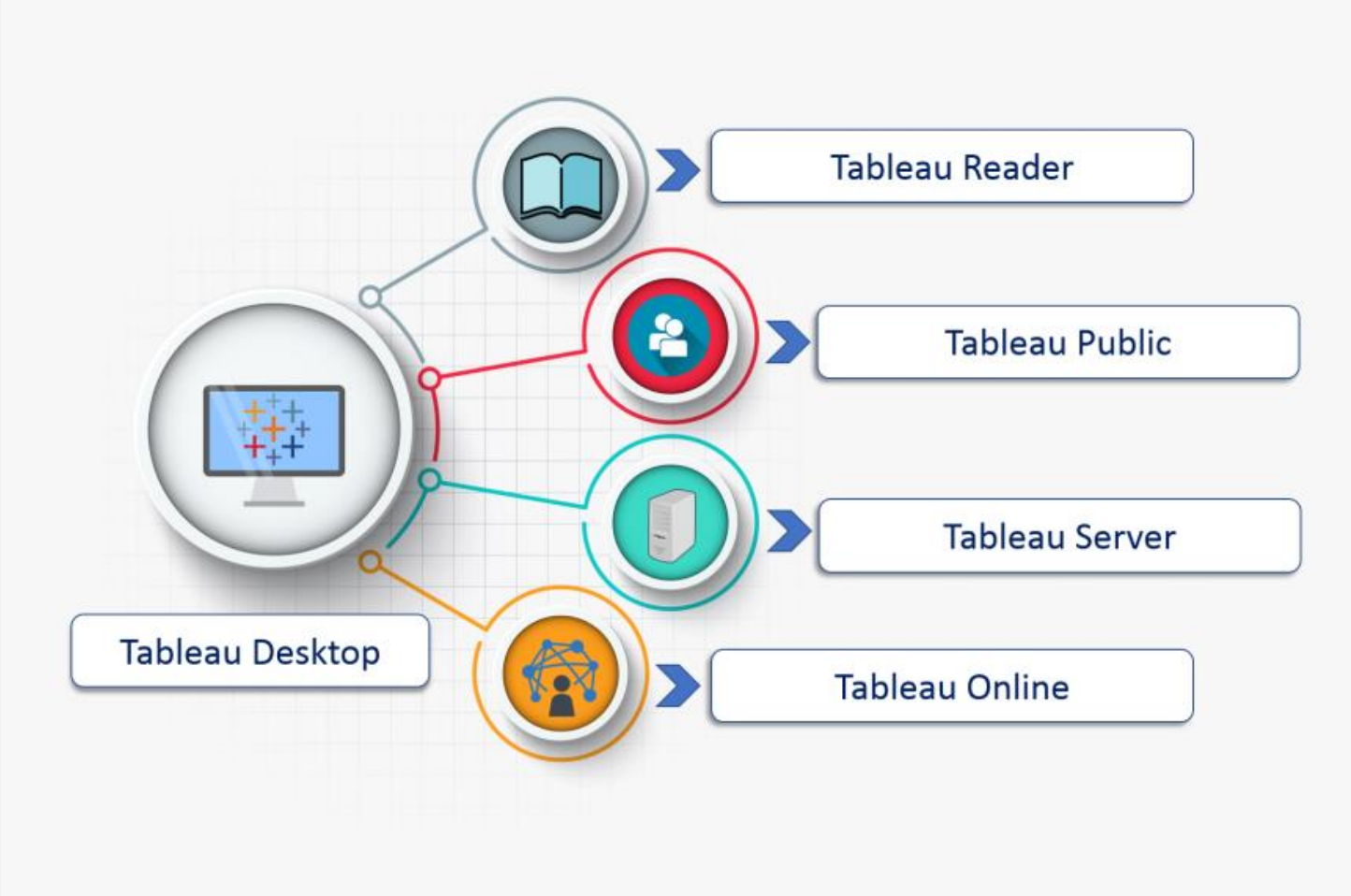
Tableau

“Tableau is a trending and market-leading BI (Business Intelligence) tool used to visualize and analyze in an easily digestible format. It allows user to work on live data-set and spend more time on data analysis rather than data wrangling.”

Tableau Advantages

- Access to Multiple Data Connection
- Live Analysis
- Interactive visual by drag and drop
- ShowMe (Graph Recommendation)
- Maps

Tableau Family Products



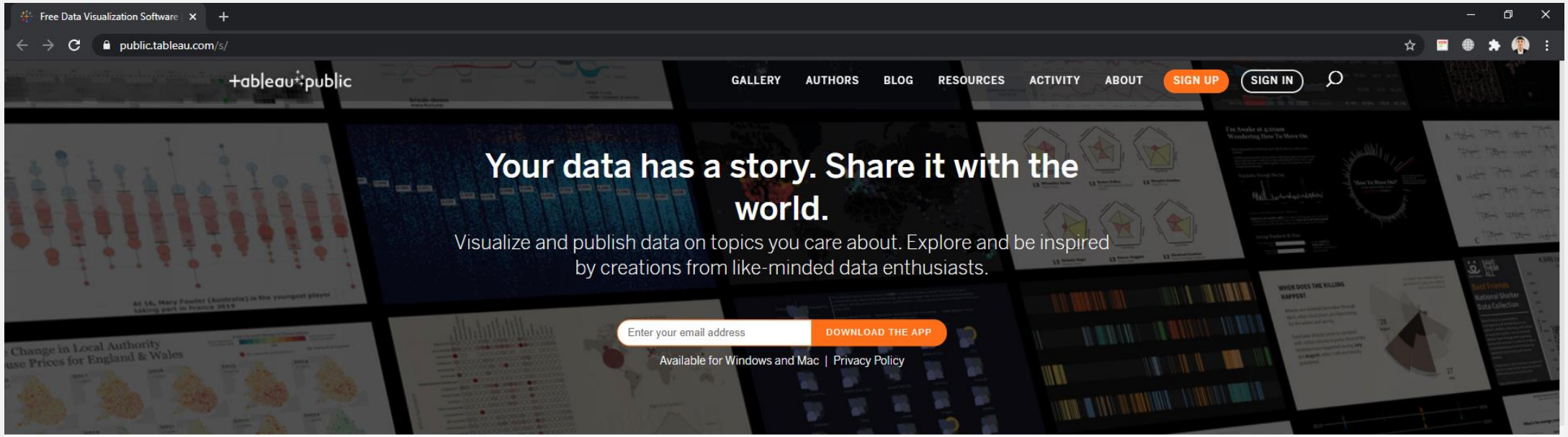
Installing Public Tableau

- As for learning purpose, we would use the **Public Tableau** version which is free for public. The website could be visited here at **https://public.tableau.com/**

- Student, ther is a possibility to gain free 1-year access for the tableau at **https://www.tableau.com/academic/students** **,** but this slide would be based on the Public Tableau. Do not worry, as the overall feature would have no difference between the public Tableau and paid one.

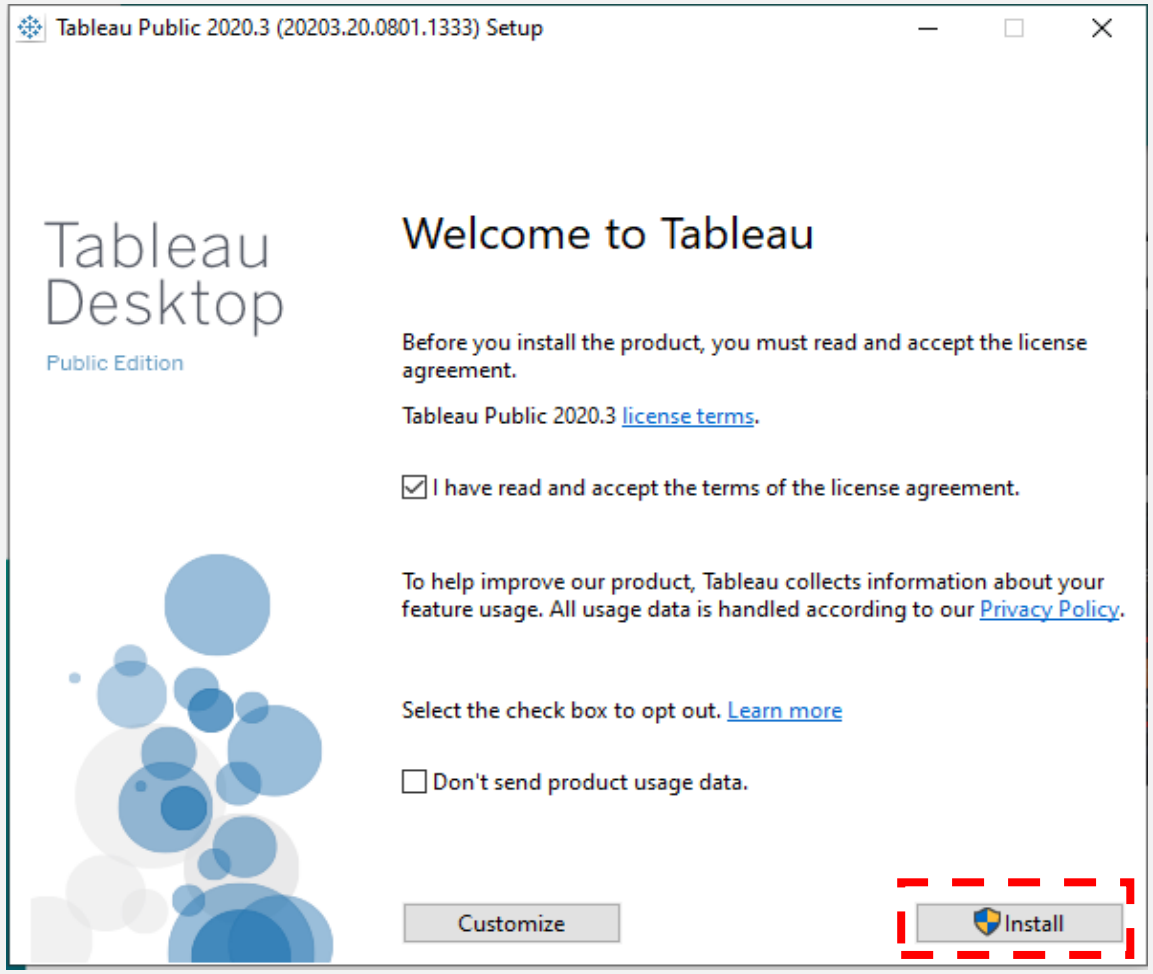
Installing Public Tableau

- Enter the E-mail address to gain access to download the software. After putting the E-mail, the application would be automatically download.



Installing Public Tableau

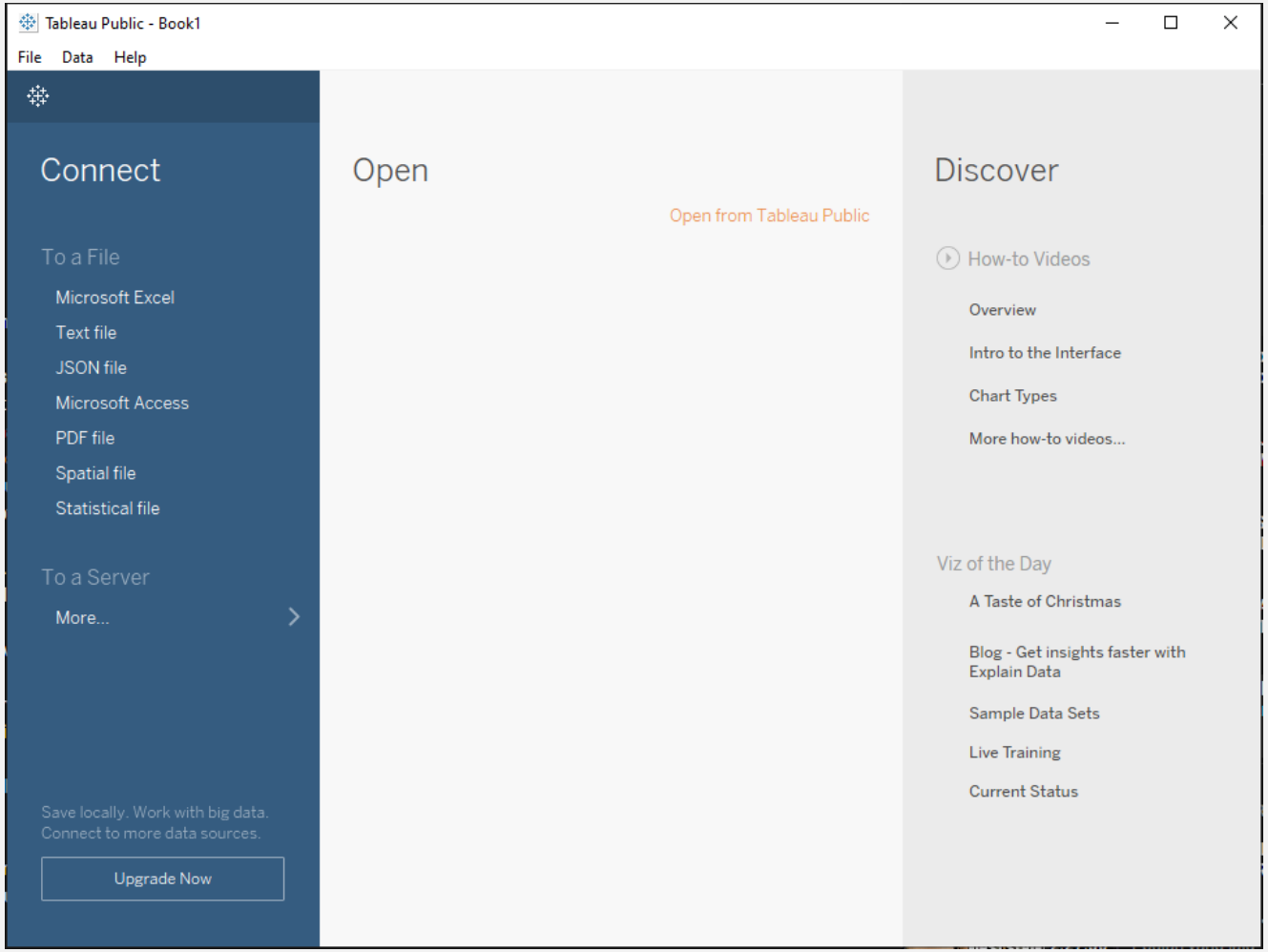
- Select Install and accept the license agreement. The process would automatically install the software. Use the customize if user want to change where the application is installed



Installing Public Tableau

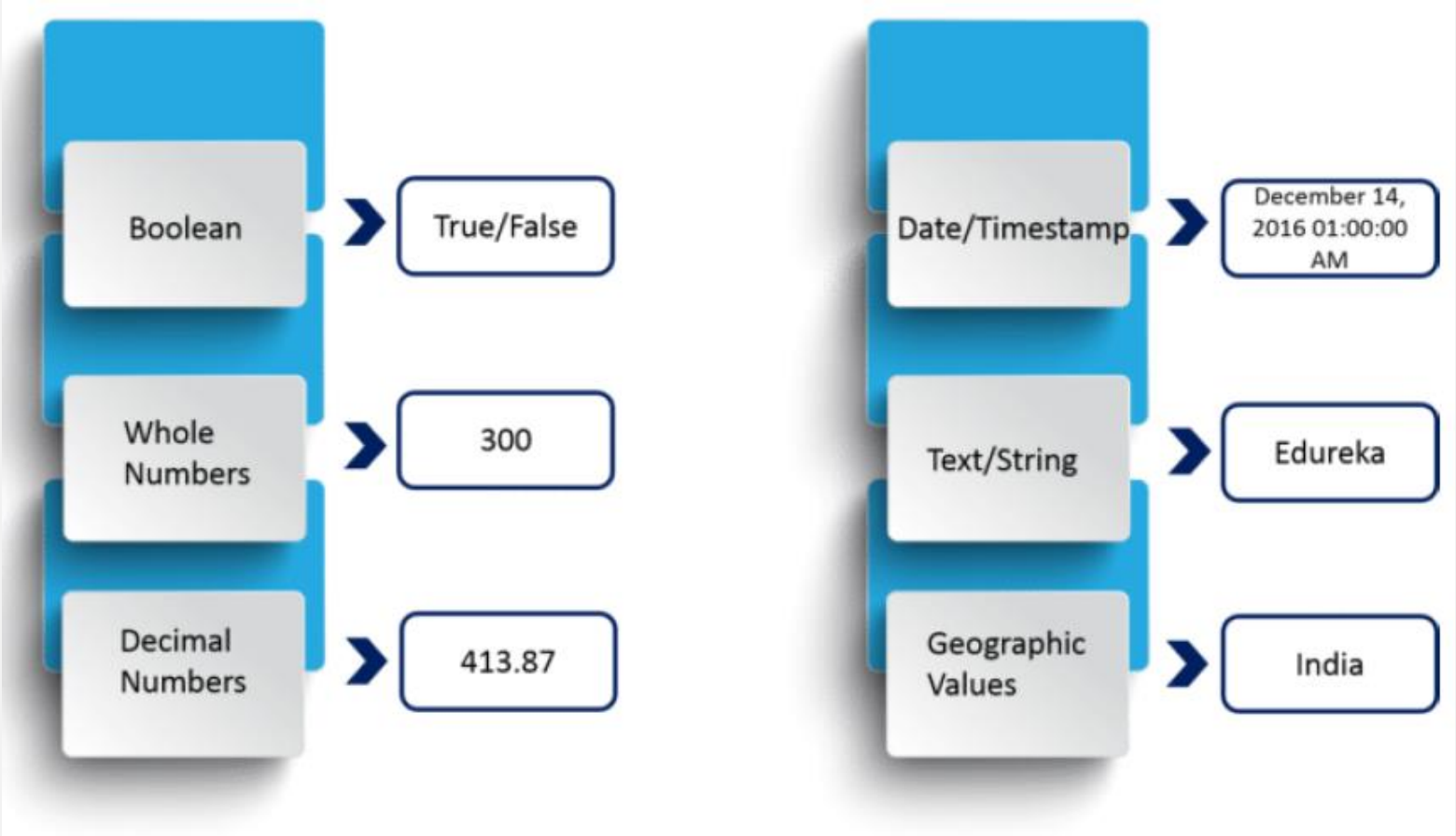
- To use tableau, first the data need to be open or in tableau term it is called Connected

- Mainly There are 2 types of connections :
  - Local server (Form File)
  - Data Server

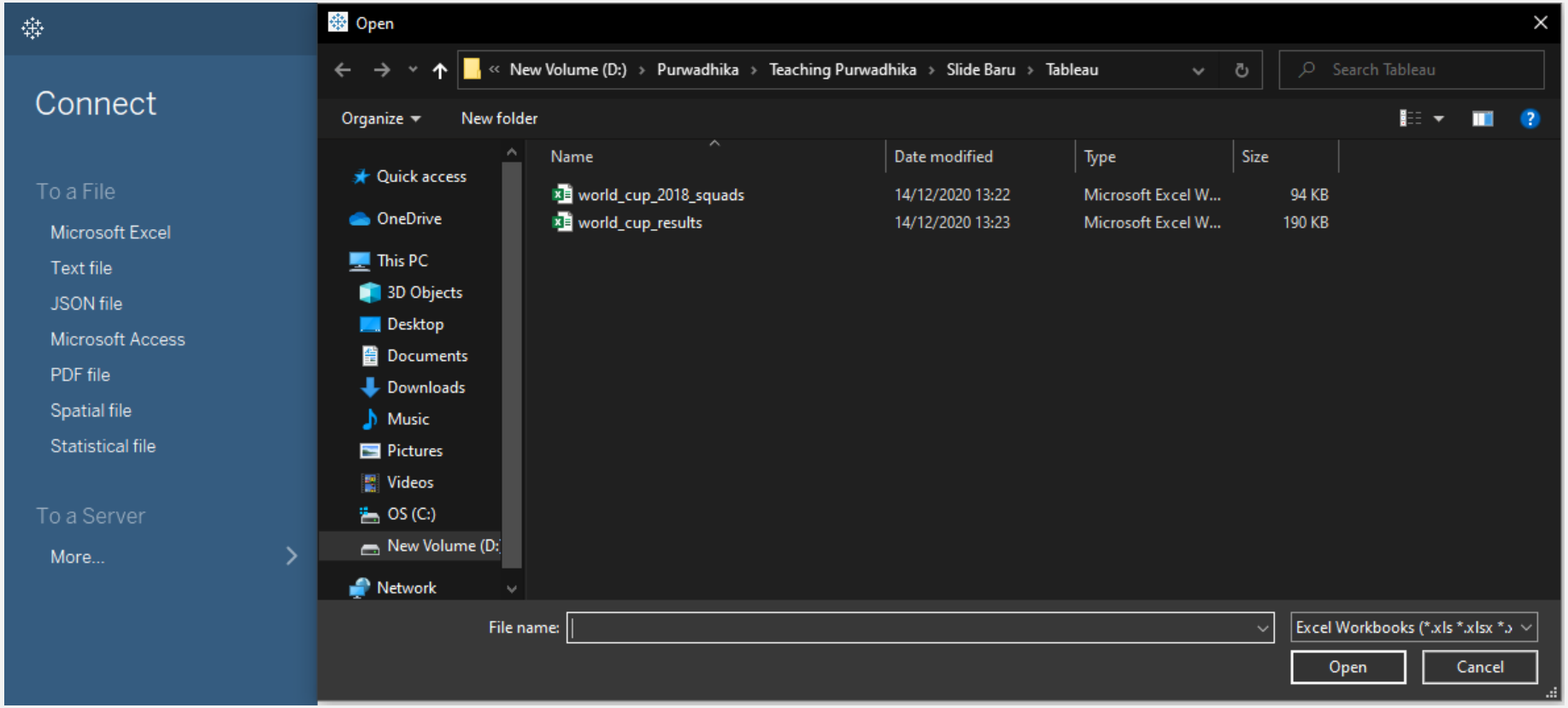


Multiple Data Connection

Tableau data types



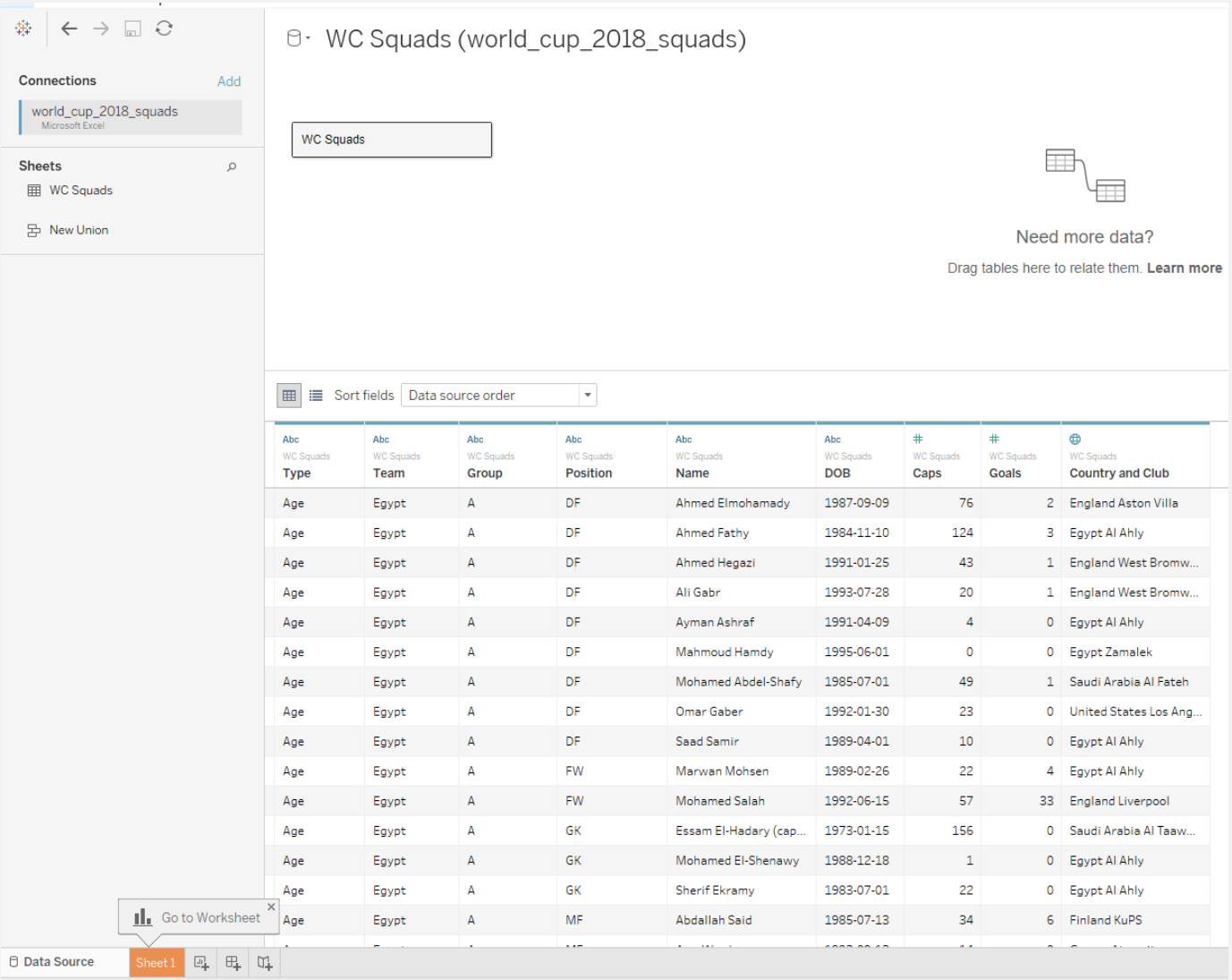
Input Dataset



Input your dataset, many files could be import to tableau (ex. Excel File, Json File, etc)

Add More Dataset

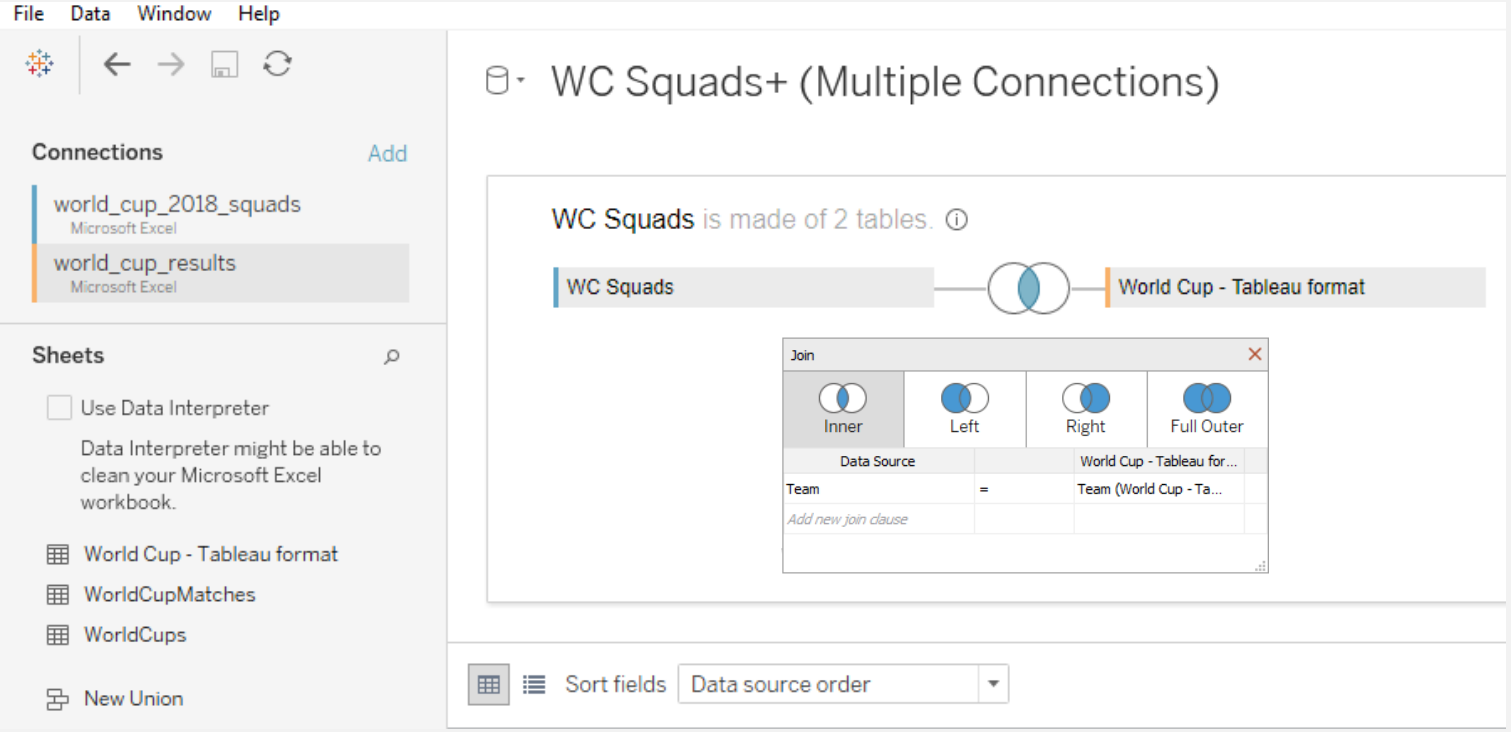
- You can add any table what you want
- Click add and direct to your file



Type	Team	Group	Position	Name	DOB	Caps	Goals	Country and Club
Age	Egypt	A	DF	Ahmed Elmohamady	1987-09-09	76	2	England Aston Villa
Age	Egypt	A	DF	Ahmed Fathy	1984-11-10	124	3	Egypt Al Ahly
Age	Egypt	A	DF	Ahmed Hegazi	1991-01-25	43	1	England West Bromw...
Age	Egypt	A	DF	Ali Gabr	1993-07-28	20	1	England West Bromw...
Age	Egypt	A	DF	Ayman Ashraf	1991-04-09	4	0	Egypt Al Ahly
Age	Egypt	A	DF	Mahmoud Hamdy	1995-06-01	0	0	Egypt Zamalek
Age	Egypt	A	DF	Mohamed Abdel-Shafy	1985-07-01	49	1	Saudi Arabia Al Fateh
Age	Egypt	A	DF	Omar Gaber	1992-01-30	23	0	United States Los Ang...
Age	Egypt	A	DF	Saad Samir	1989-04-01	10	0	Egypt Al Ahly
Age	Egypt	A	FW	Marwan Mohsen	1989-02-26	22	4	Egypt Al Ahly
Age	Egypt	A	FW	Mohamed Salah	1992-06-15	57	33	England Liverpool
Age	Egypt	A	GK	Essam El-Hadary (cap...	1973-01-15	156	0	Saudi Arabia Al Taaw...
Age	Egypt	A	GK	Mohamed El-Shenawy	1988-12-18	1	0	Egypt Al Ahly
Age	Egypt	A	GK	Sherif Ekramy	1983-07-01	22	0	Egypt Al Ahly
Age	Egypt	A	MF	Abdallah Said	1985-07-13	34	6	Finland KuPS

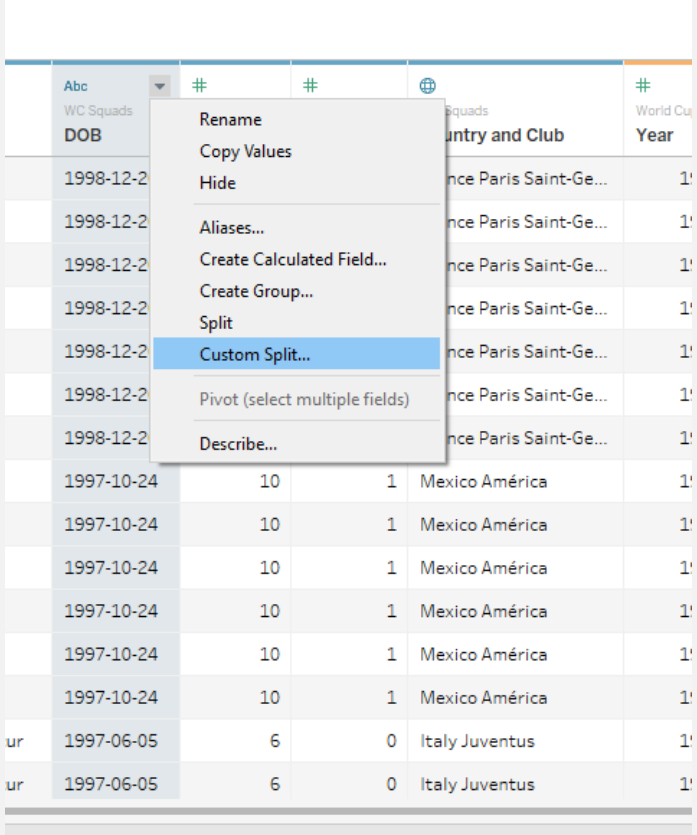
Data Joins

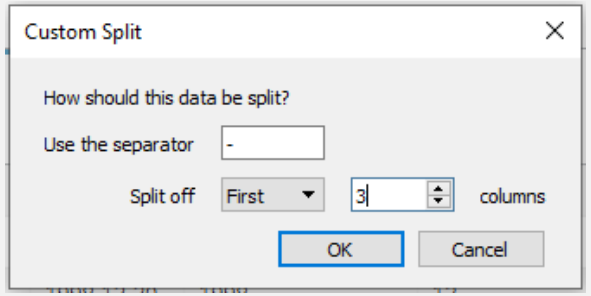
- Joining data in tableau just drop the table what you want to join with another table.
- Type of join in tableau (Inner, Left, Right and Full Outer)



Custom Split

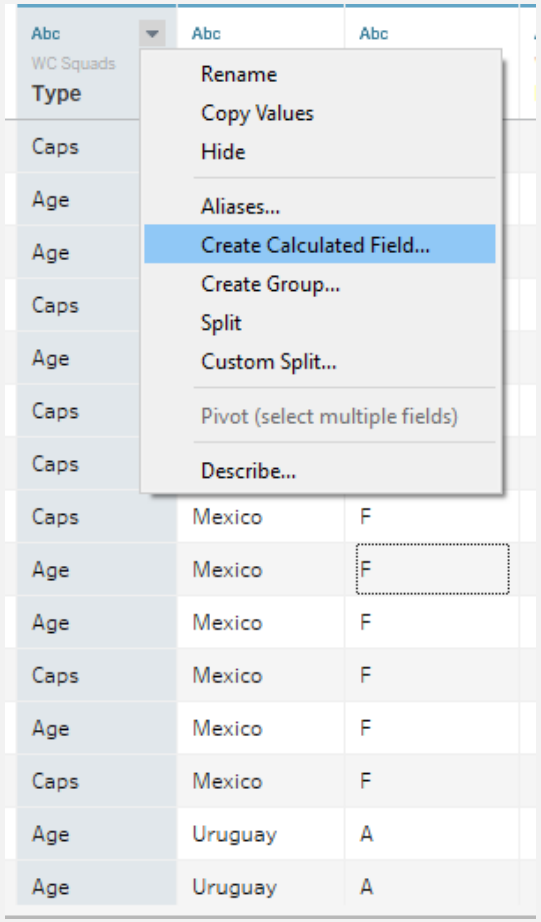
- With custom split you can breakdown columns **DOB** to 3 parts, years, month and date.





Abc WC Squads DOB	=Abc Calculation DOB - Split 1	=Abc Calculation DOB - Split 2	=Abc Calculation DOB - Split 3
1998-12-20	1998	12	20
1998-12-20	1998	12	20
1998-12-20	1998	12	20
1998-12-20	1998	12	20
1998-12-20	1998	12	20
1998-12-20	1998	12	20
1998-12-20	1998	12	20
1997-10-24	1997	10	24
1997-10-24	1997	10	24
1997-10-24	1997	10	24
1997-10-24	1997	10	24
1997-10-24	1997	10	24
1997-10-24	1997	10	24
1997-06-05	1997	06	05
1997-06-05	1997	06	05

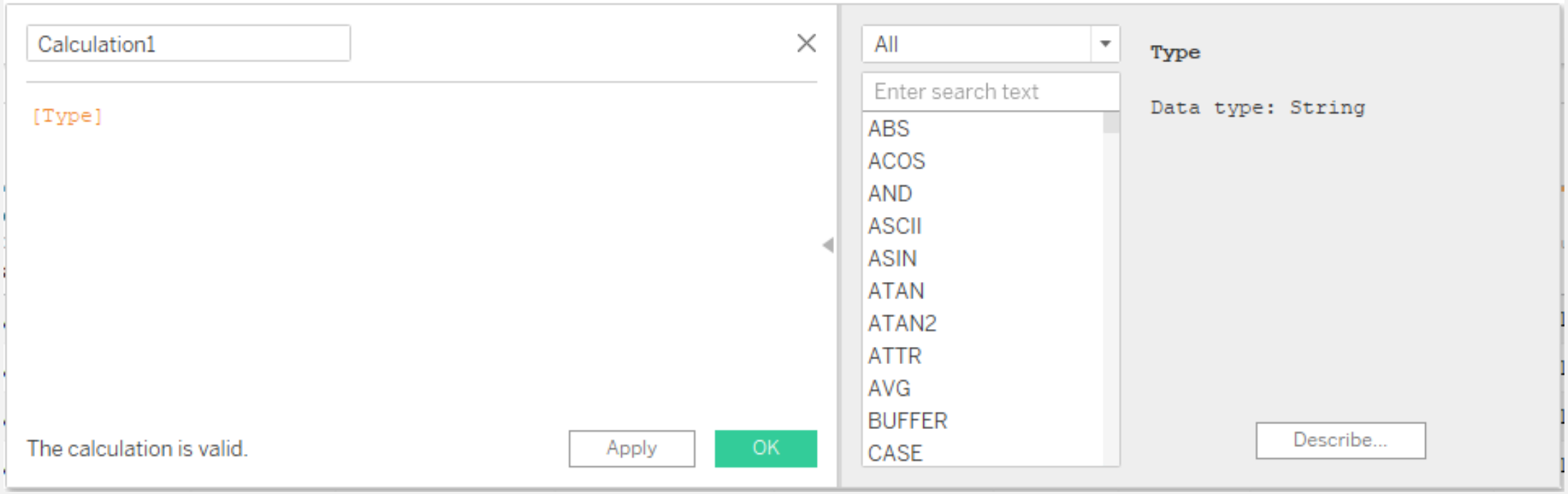
Data Calculation



- Tableau support the data calculation with similar syntax SQL or Excel
- The calculation function is useful to manipulate the data in order to get better insight
- Types of operator supported at Tableau are :
  - General Operators
  - Arithmetic Operators
  - Relational Operators
  - Logical Operators
- Different Categories of function at Tableau are :
  - Number Functions
  - String Functions
  - Date Functions
  - Logical Functions
  - Aggregate Functions

Data Calculation

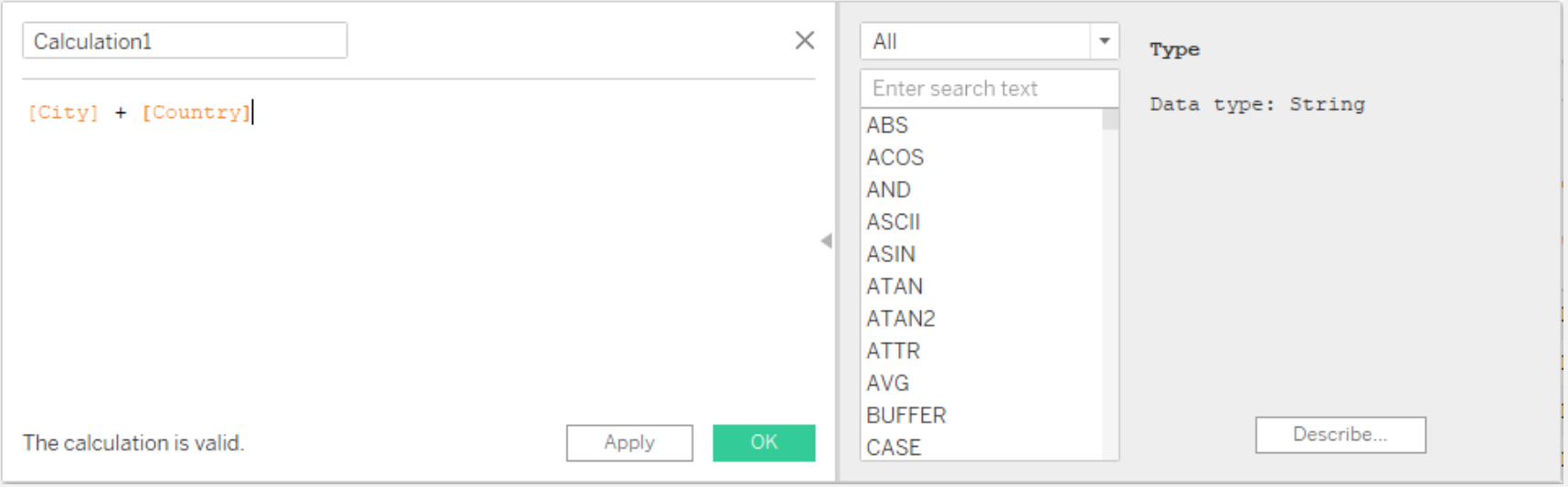
- The calculation field should be shown up like this. If the calculation opened from the column, the column name would be shown in the calculation field.



General Operators

- Following table shows the general operators supported by Tableau. These operators act on numeric, character and date data types.

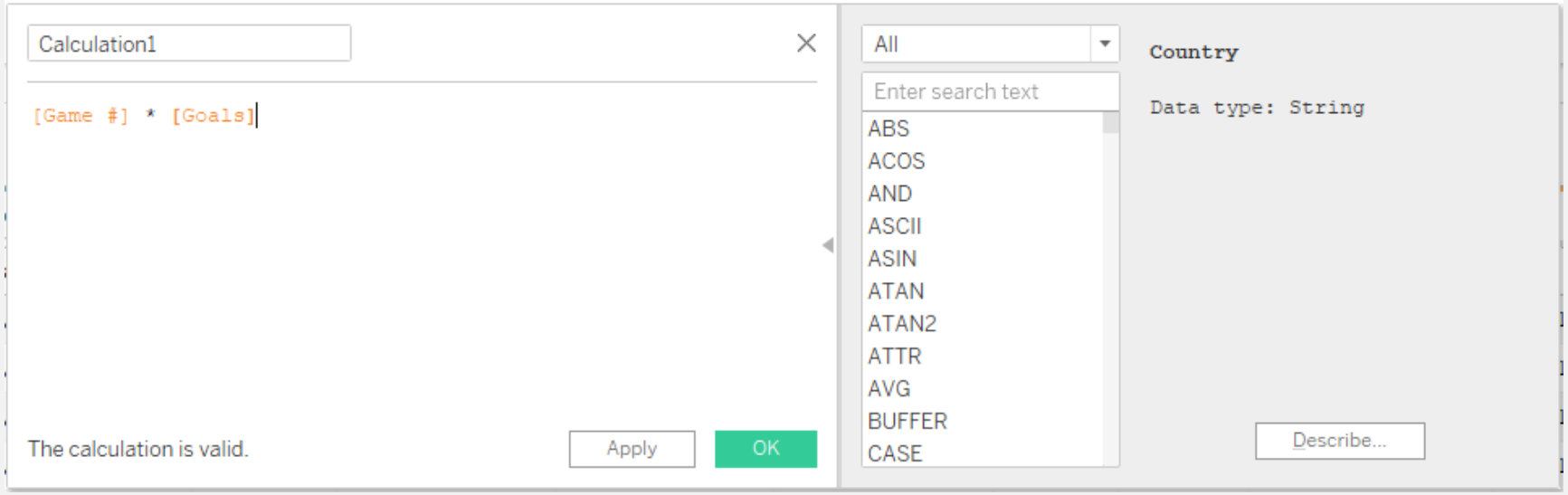
Operator	Description
+	Adds two numbers. Concatenates two strings. Adds days to dates.
–	Subtracts two numbers. Subtracts days from dates.



Arithmetic Operators

- These operators act only on numeric data types.

Operator	Description
*	Numeric multiplication
/	Numeric division
%	Reminder of numeric division
^	Raised to the power



- Both of the columns are numeric. This made the arithmetic operation work

Relational Operators

- These operators are used in expressions. Each operators compares two numbers, dates, or strings and returns a Boolean (**TRUE of FALSE**). Booleans themselves, however, cannot be compared using these operators

Operator	Description
= = or = (Equal to)	Compares two numbers or two strings or two dates to be equal. Returns the Boolean value TRUE if they are, else returns false.
!= or <> (Not equal to)	Compares two numbers or two strings or two dates to be unequal. Returns the Boolean value TRUE if they are, else returns false.
> (Greater than)	Compares two numbers or two strings or two dates where the first argument is greater than second. Returns the Boolean value TRUE if it is the case, else returns false.
< (Less than)	Compares two numbers or two strings or two dates where the first argument is smaller than second. Returns the Boolean value TRUE if it is the case, else returns false.

Logical Operators

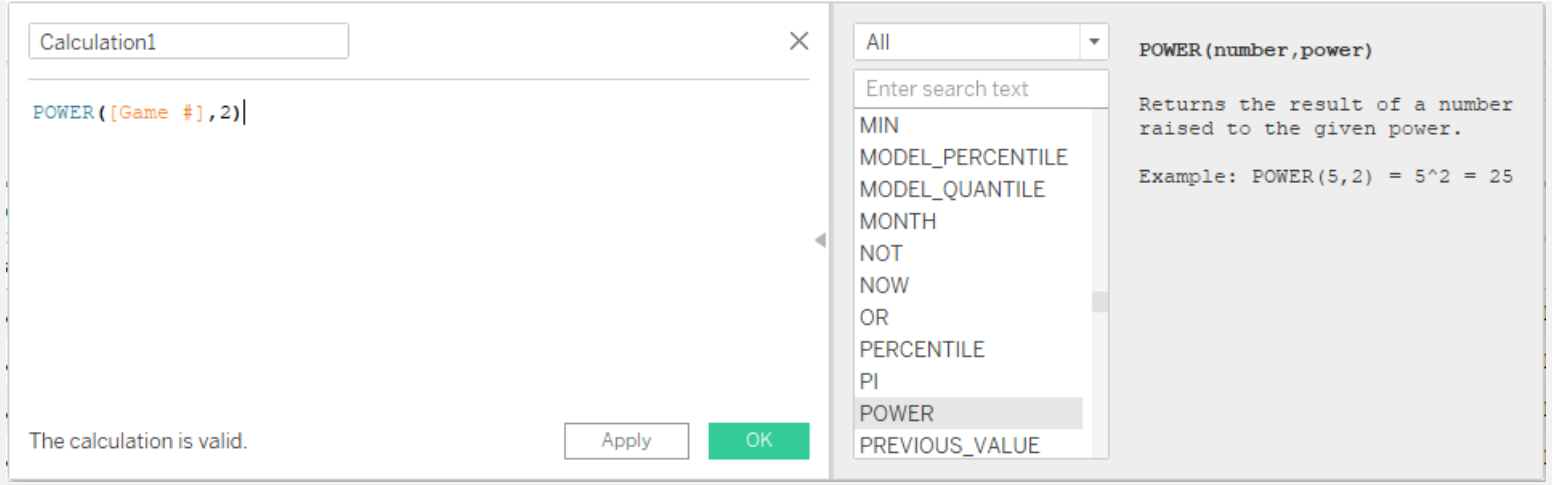
- These operators are used in expressions whose result is a Boolean giving the output as TRUE or FALSE.

Operator	Description
AND	If the expressions or Boolean values present on both sides of AND operator is evaluated to be TRUE, then the result is TRUE. Else the result is FALSE.
OR	If any one or both of the expressions or Boolean values present on both sides of AND operator is evaluated to be TRUE, then the result is TRUE. Else the result is FALSE.
NOT	This operator negates the Boolean value of the expression present after it.

Number Functions

- These are the functions used for numeric calculations. They only take number as inputs. Following are some examples of important number functions.

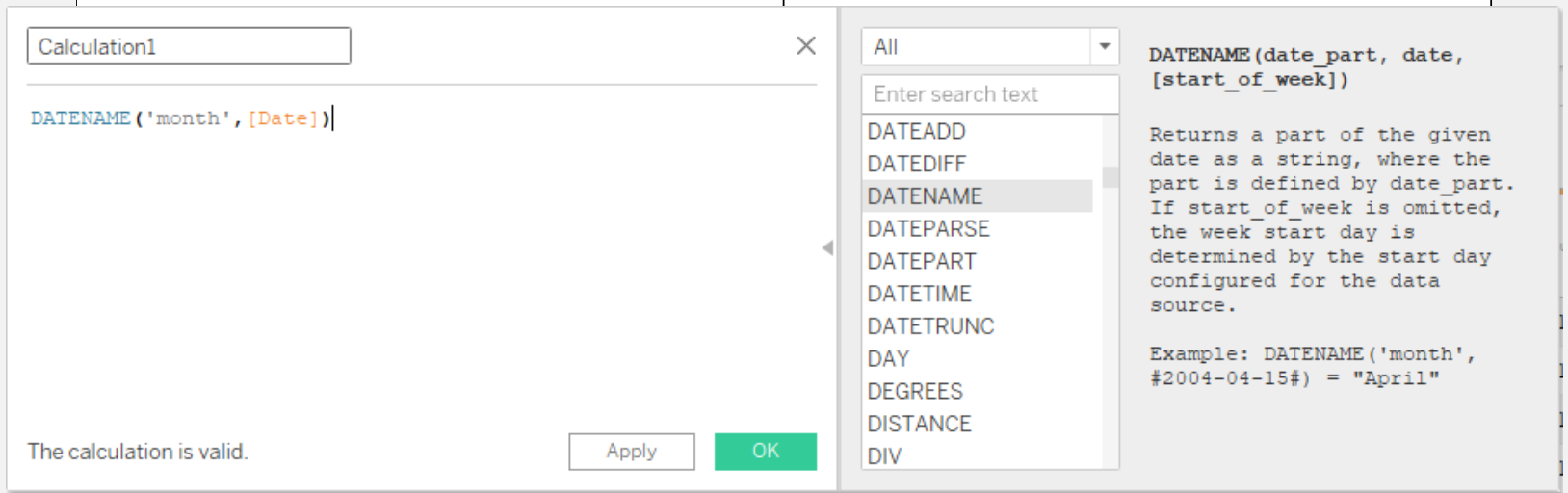
Function	Description
CEILING (number)	Rounds a number to the nearest integer of equal or greater value.
POWER (number, power)	Raises the number to the specified power.
ROUND (number, [decimals])	Rounds the numbers to a specified number of digits.



Date Functions

- All the date functions use the **date_part** which is a string indicating the part of the date such as, year, month or day. Following this table list some examples of important date functions.

Function	Description
DATEADD (date_part, increment, date)	Returns an increment added to the date. The type of increment is specified in **date_part**.
DATENAME (date_part, date, [start_of_week])	Returns **date_part** of date as a string. The **start_of_week**parameter is optional.
DAY (date)	Returns the day of the given date as an integer.
NOW( )	Returns the current date and time.



Logical Functions

- These functions evaluate some single value or the result of an expression and produce a Boolean output.

Function	Description
IFNULL (expression1, expression2)	The IFNULL function returns the first expression if the result is not null, and returns the second expression if it is null.
ISDATE (string)	The ISDATE function returns TRUE if the string argument can be converted to a date, and FALSE if it cannot.
MIN(expression)	The MIN function returns the minimum of an expression across all records or the minimum of two expressions for each record.

Aggregate Functions

- Aggregate Functions are a type of function where values of multiple rows are grouped together as the input to from a single value of more significant meaning, such as a set or list. Following table are some examples of the Aggregate function.

Function	Description
AVG(expression)	Returns the average of all the values in the expression. AVG can be used with numeric fields only. Null values are ignored.
COUNT (expression)	Returns the number of items in a group. Null values are not counted.
MEDIAN (expression)	Returns the median of an expression across all records. Median can only be used with numeric fields. Null values are ignored.
STDEV (expression)	Returns the statistical standard deviation of all values in the given expression based on a sample of the population.

Data Visualization

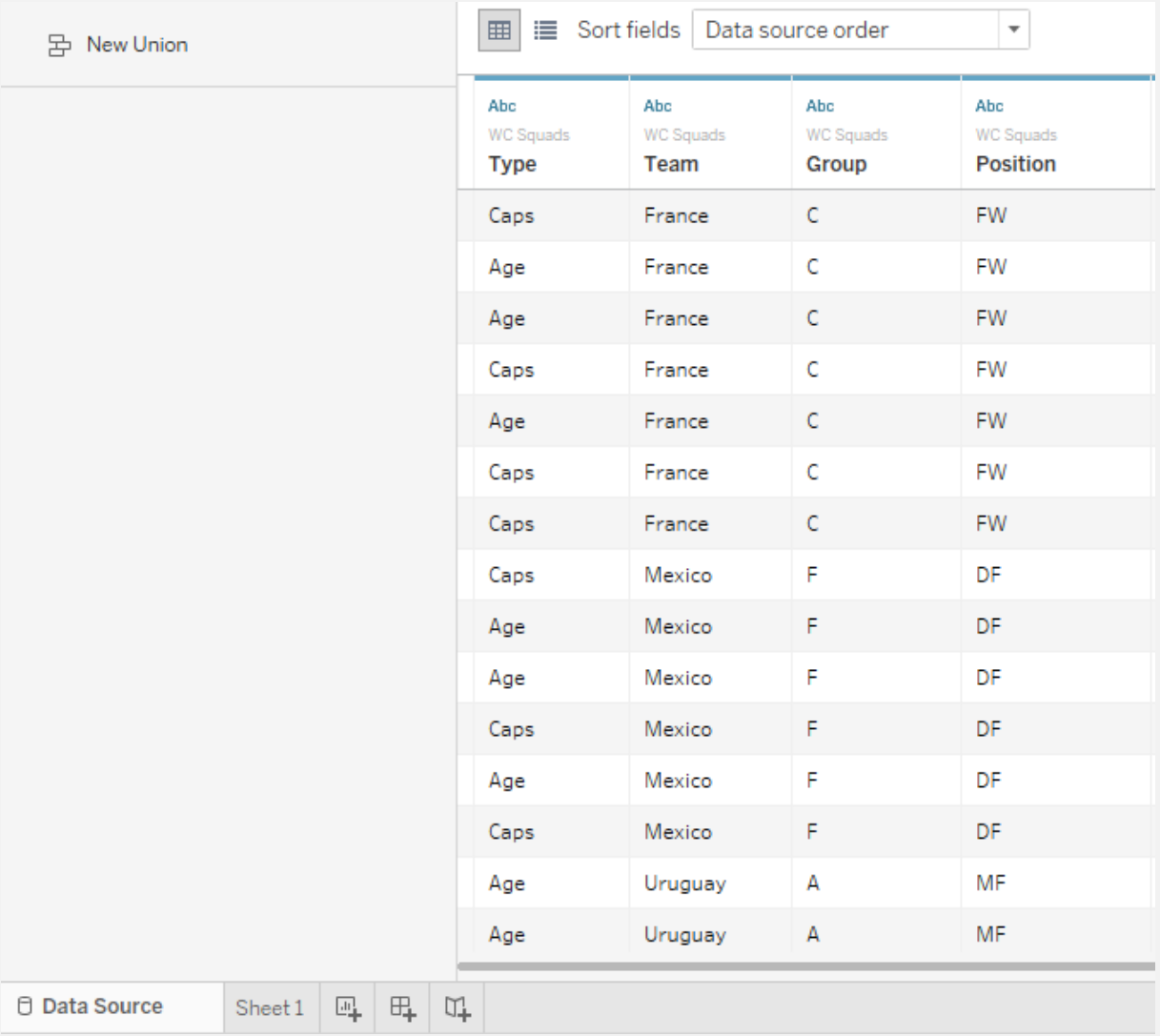
Tableau main feature is the Data visualization tool that is widely used for Business Intelligence. It helps create interactive graphs and charts in the form of dashboard and worksheets to gain business insight. And all of this is made possible with gestures as simple as drag and drop!

Graph Examples

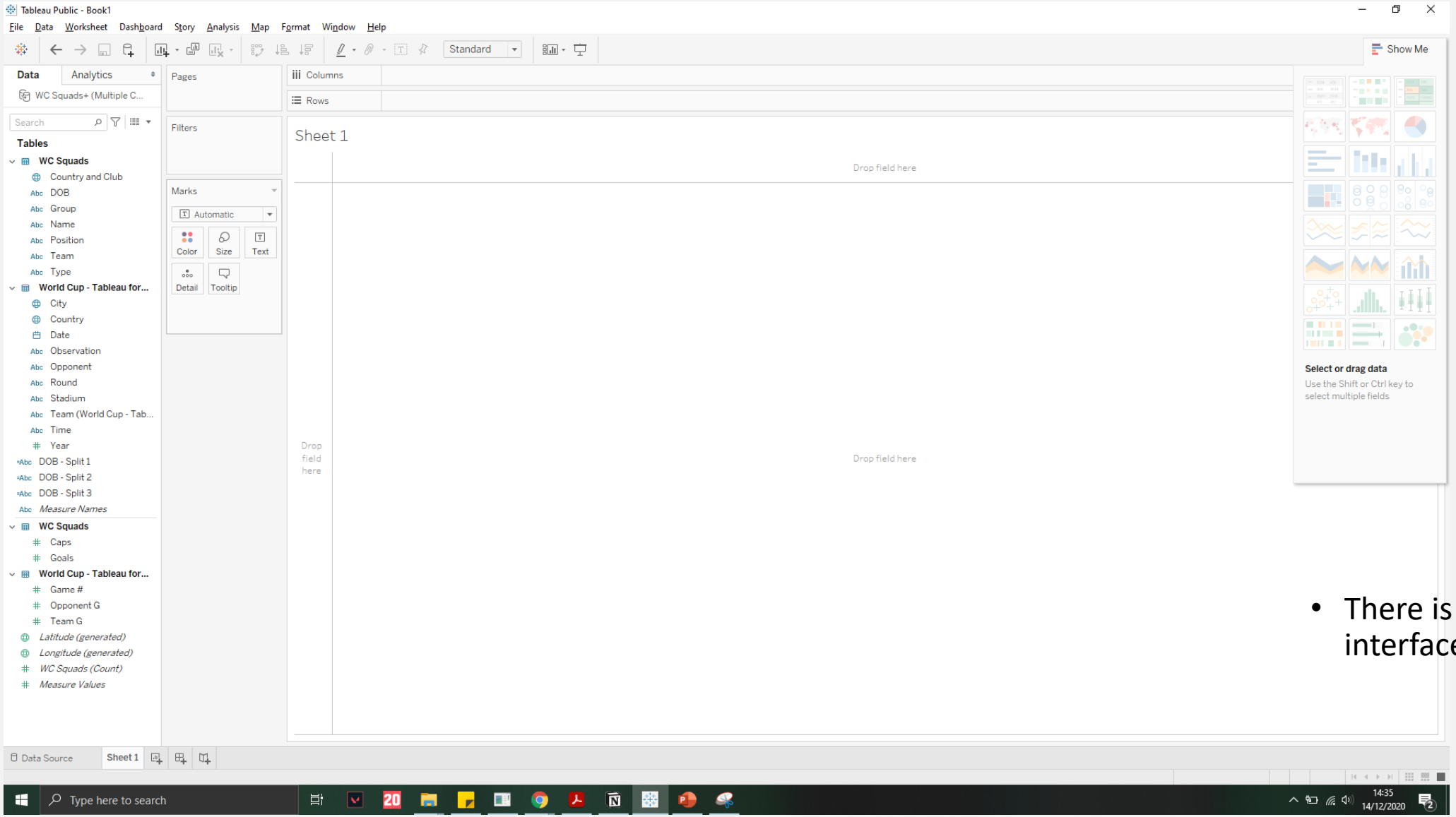
Visualization	Application
Bar Graph	Used when Dimension is discontinuous
Line Graph	Preferred for continuous Dimensions
Dual Axis Graph	Used to represent two Measures together
Geographical Graph	Used to plot Measures on geographical map
Area Graph – Dual Axes	Provides better comparison amongst Measures
Heat Map	Used to visualize variations across categories
Tree Map	Used to represent quantity in nested rectangles

Data Visualization

- As soon as data imported, select worhksheet to create data visualization
- You can add the sheet like an excel sheet, can be change (rename)



Abc WC Squads Type	Abc WC Squads Team	Abc WC Squads Group	Abc WC Squads Position
Caps	France	C	FW
Age	France	C	FW
Age	France	C	FW
Caps	France	C	FW
Age	France	C	FW
Caps	France	C	FW
Caps	France	C	FW
Caps	Mexico	F	DF
Age	Mexico	F	DF
Age	Mexico	F	DF
Caps	Mexico	F	DF
Age	Mexico	F	DF
Caps	Mexico	F	DF
Age	Uruguay	A	MF
Age	Uruguay	A	MF



- There is worksheets interfaces in generally.

Tableau data type term

Dimensions

Abc

A Dimension is a field that is an independent variable.

A dimension is usually text

Measures

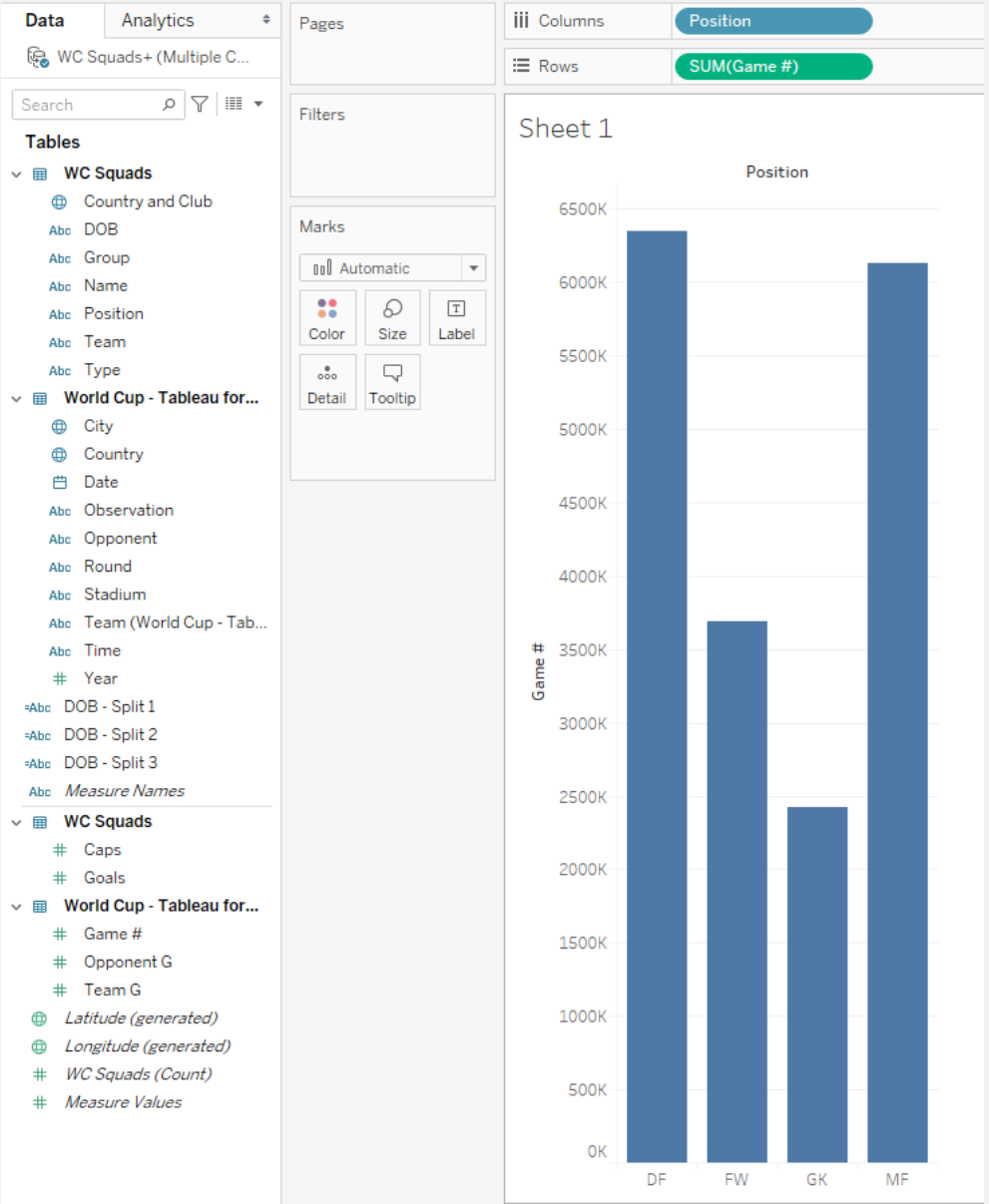
#

A Measure is a field that is a Dependent Variable and its value is a function of one or more Dimensions.

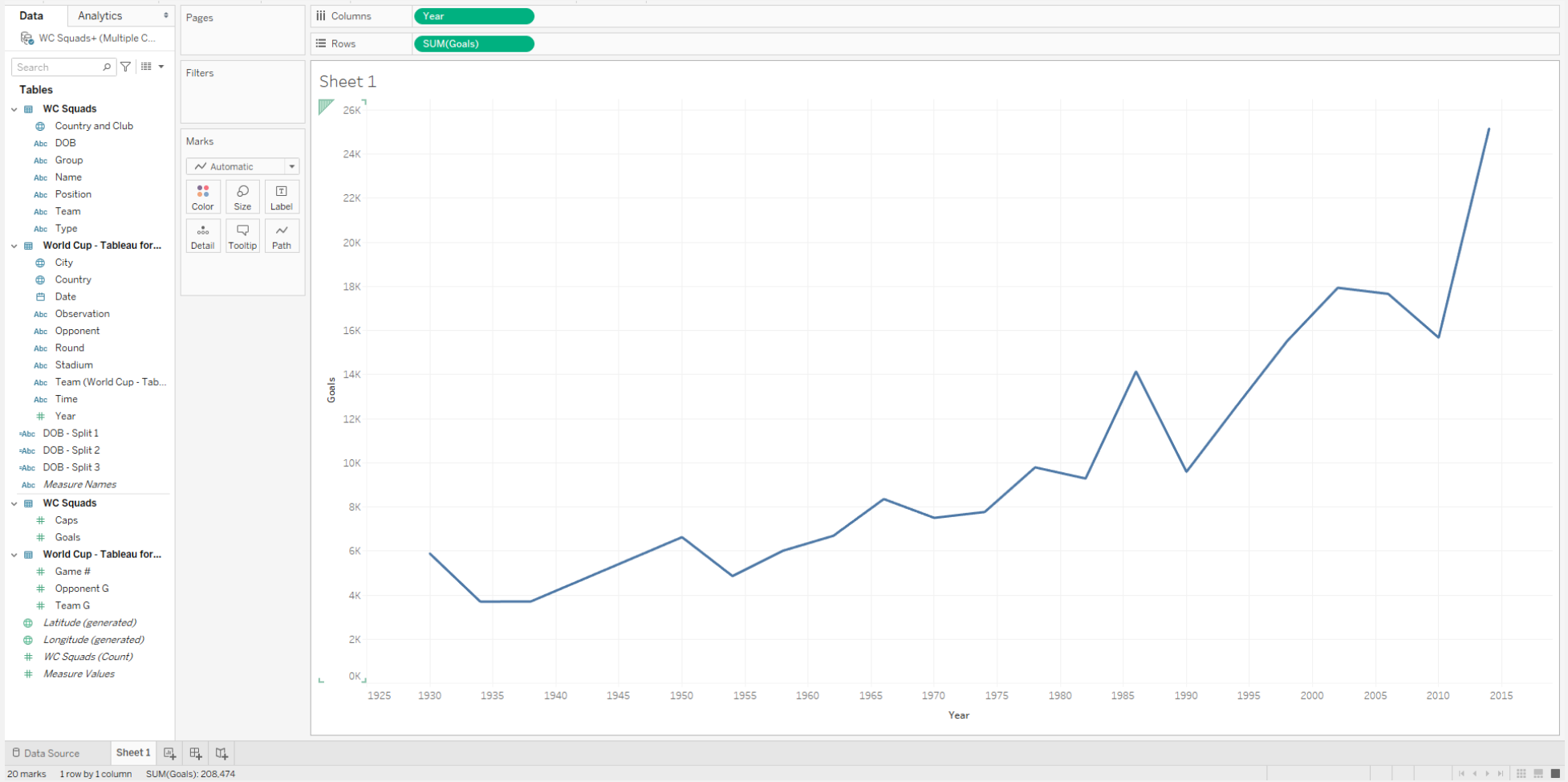
A measure is usually a number

Bar Graph

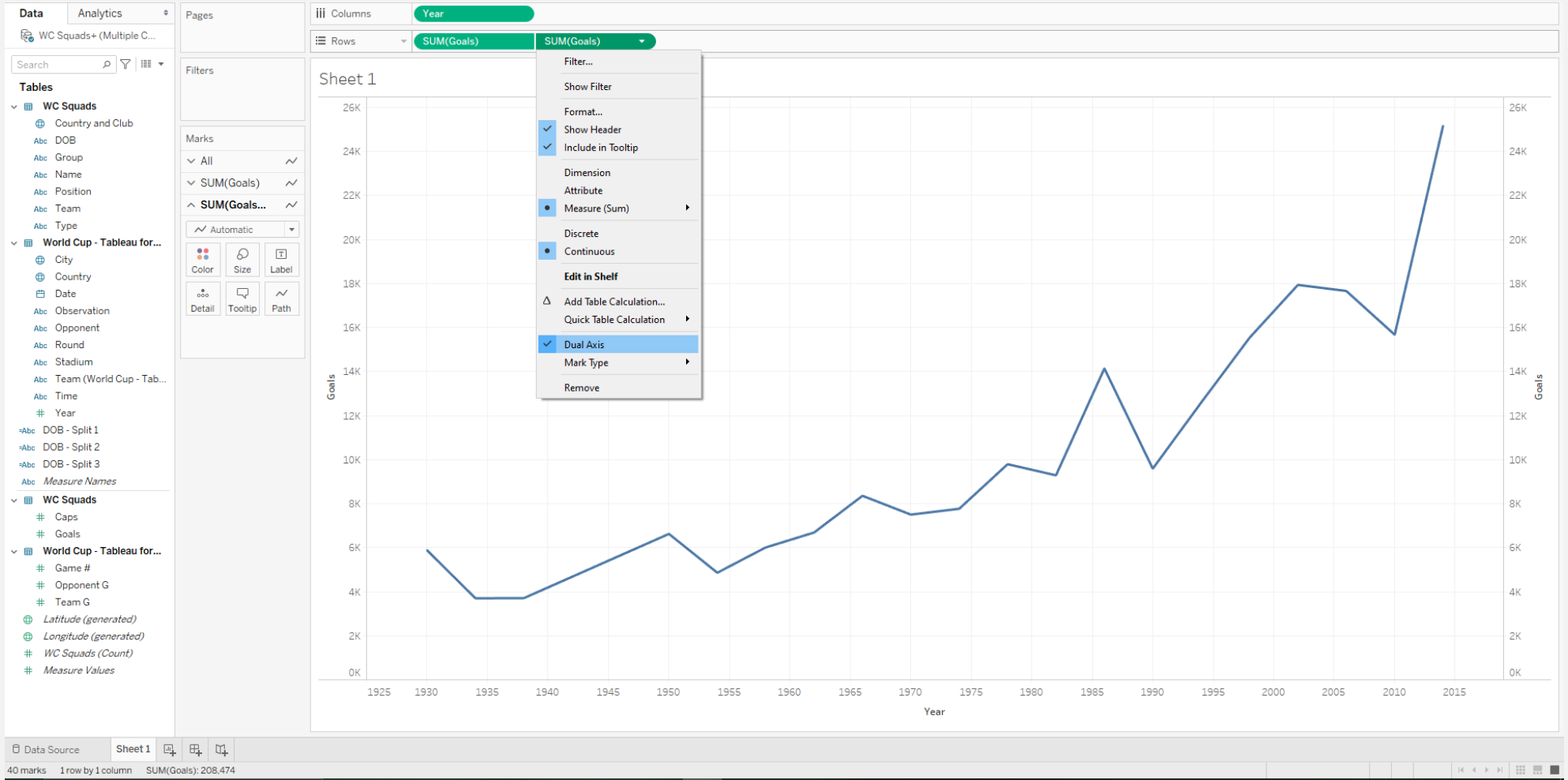
- Drag the position into columns and Game# into the rows
- The bar graph would be created automatically. If it is not, go to the Marks and select the Bar graph



Line Graph

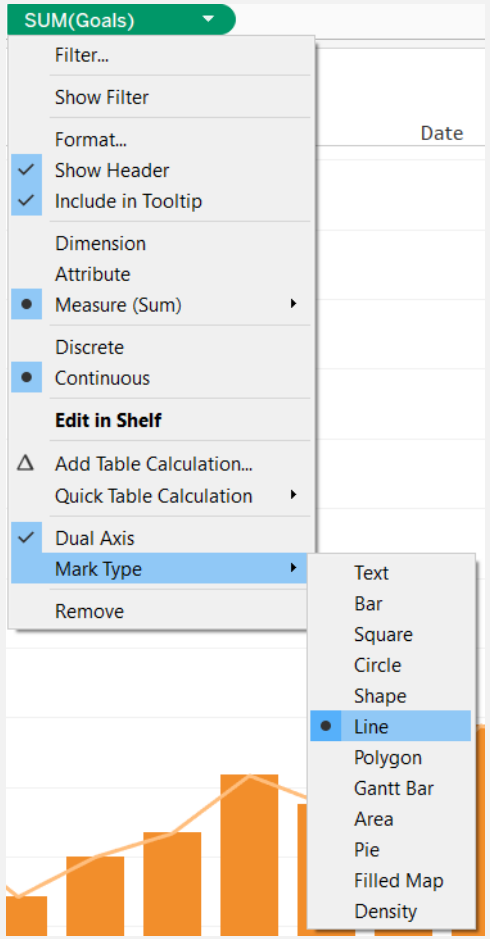

Data
Analytics
WC Squads+ (Multiple C...
Search
Tables
WC Squads
Country and Club
DOB
Group
Name
Position
Team
Type
World Cup - Tableau for...
City
Country
Date
Observation
Opponent
Round
Stadium
Team (World Cup - Tab...
Time
Year
DOB - Split 1
DOB - Split 2
DOB - Split 3
Measure Names
WC Squads
Caps
Goals
World Cup - Tableau for...
Game #
Opponent G
Team G
Latitude (generated)
Longitude (generated)
WC Squads (Count)
Measure Values
Pages
Filters
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Path
Columns
Year
Rows
SUM(Goals)
Sheet 1
26K
24K
22K
20K
18K
16K
14K
12K
10K
8K
6K
4K
2K
0K
Goals
1925
1930
1935
1940
1945
1950
1955
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
Year
Data Source
Sheet 1
20 marks
1 row by 1 column
SUM(Goals): 208,474

Dual Axis Graph

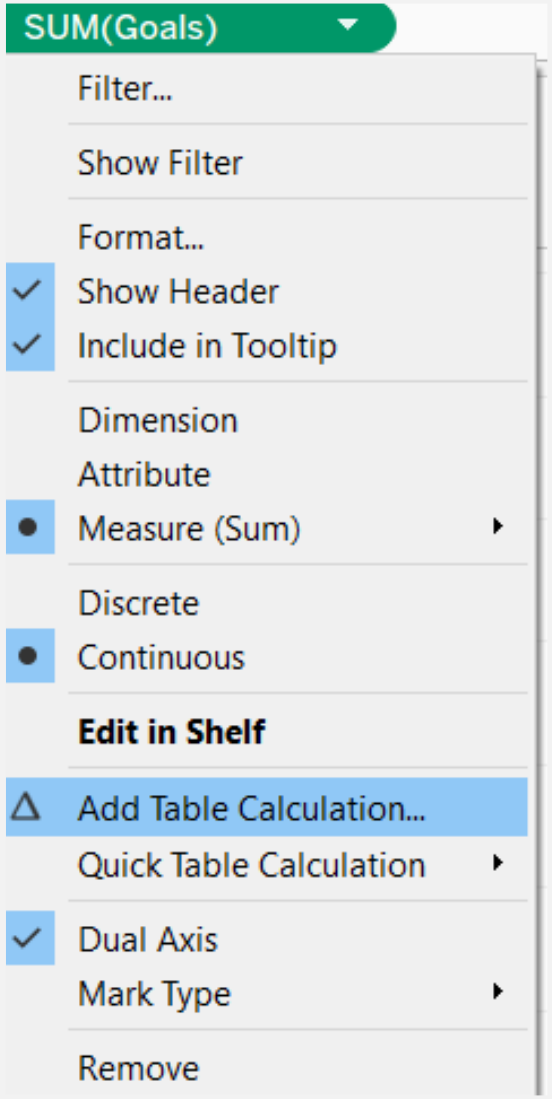


Dual Axis Graph

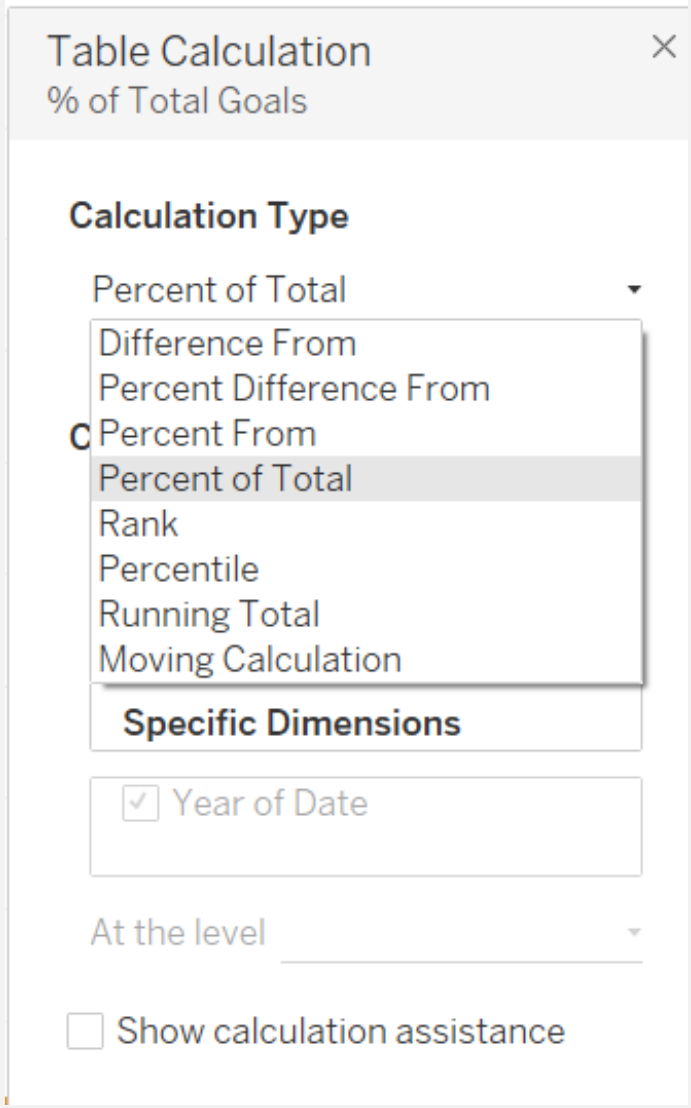
1



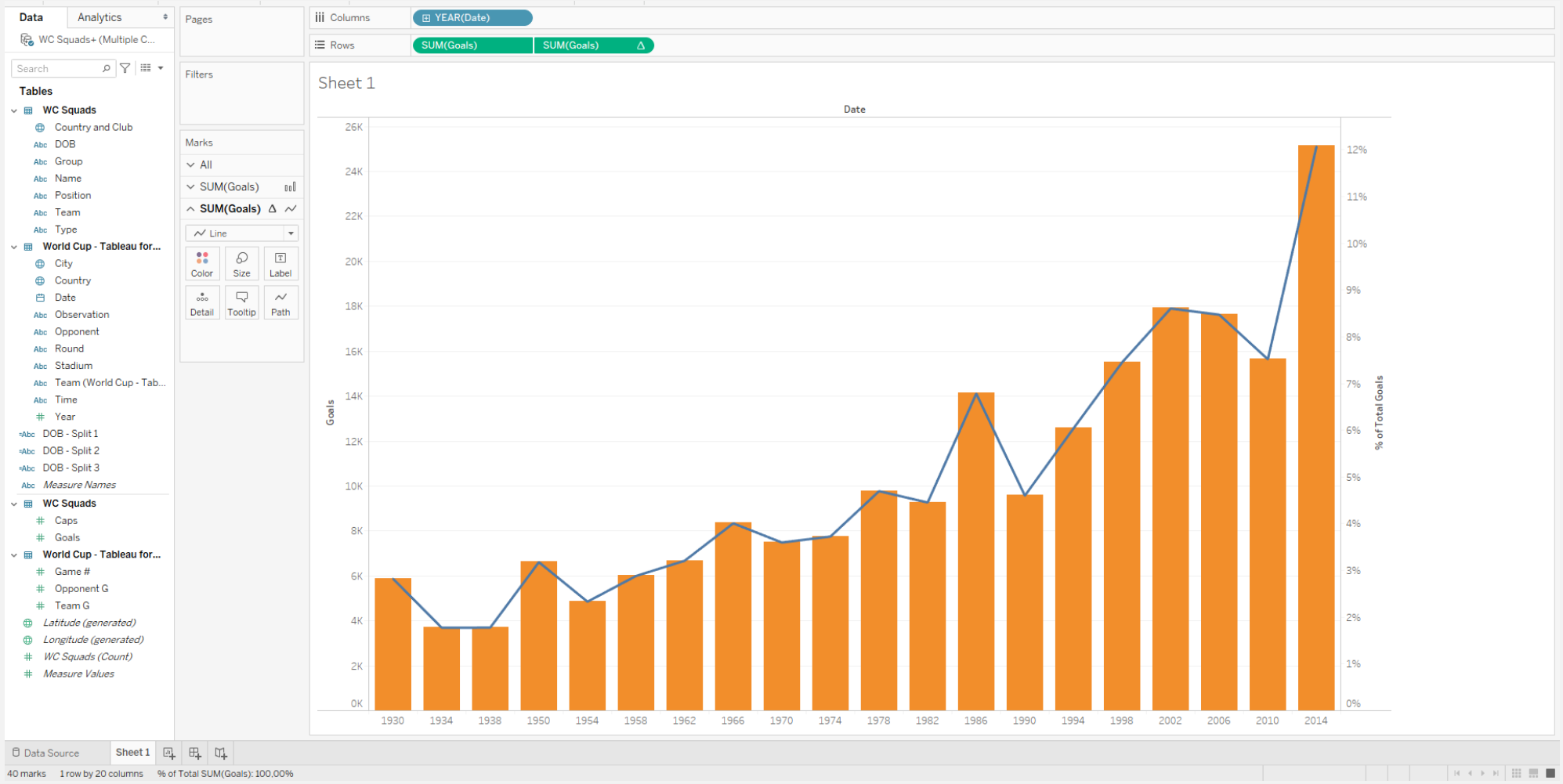
2



3

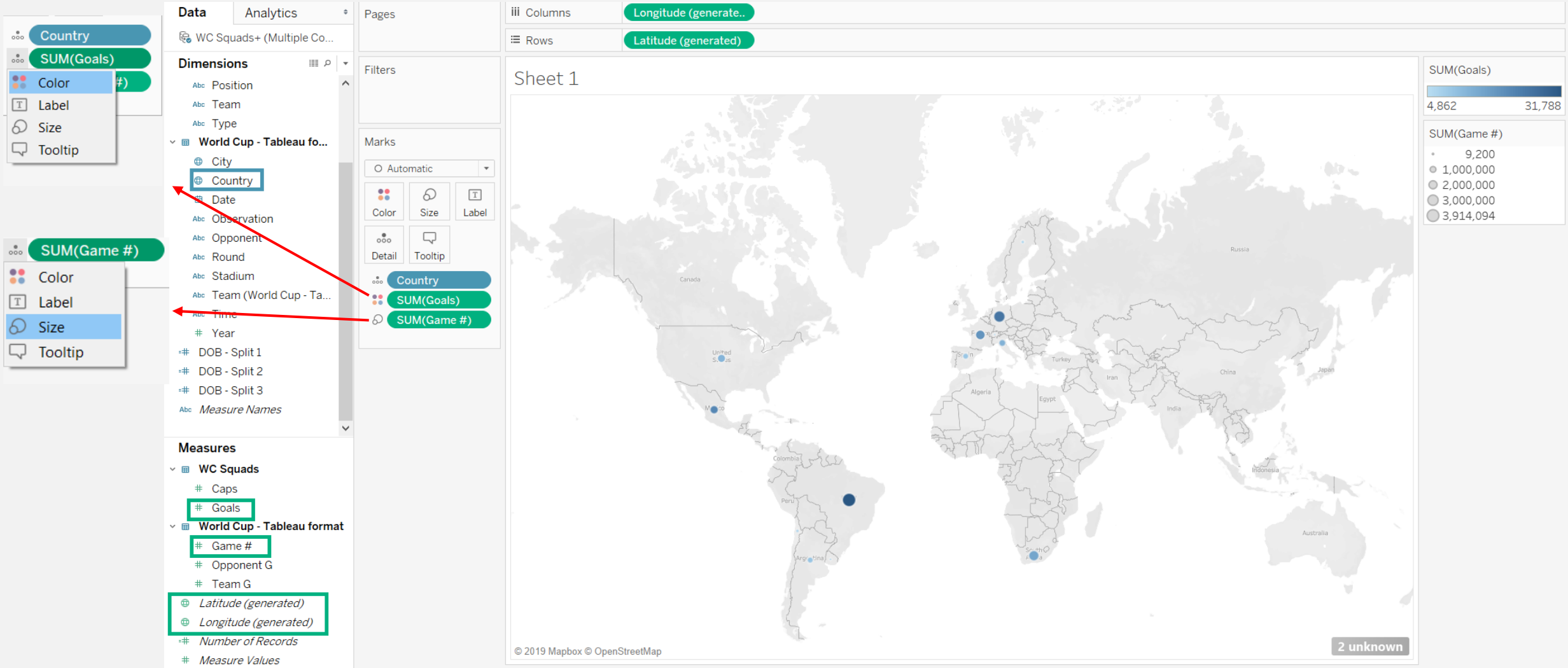


Dual Axis Graph

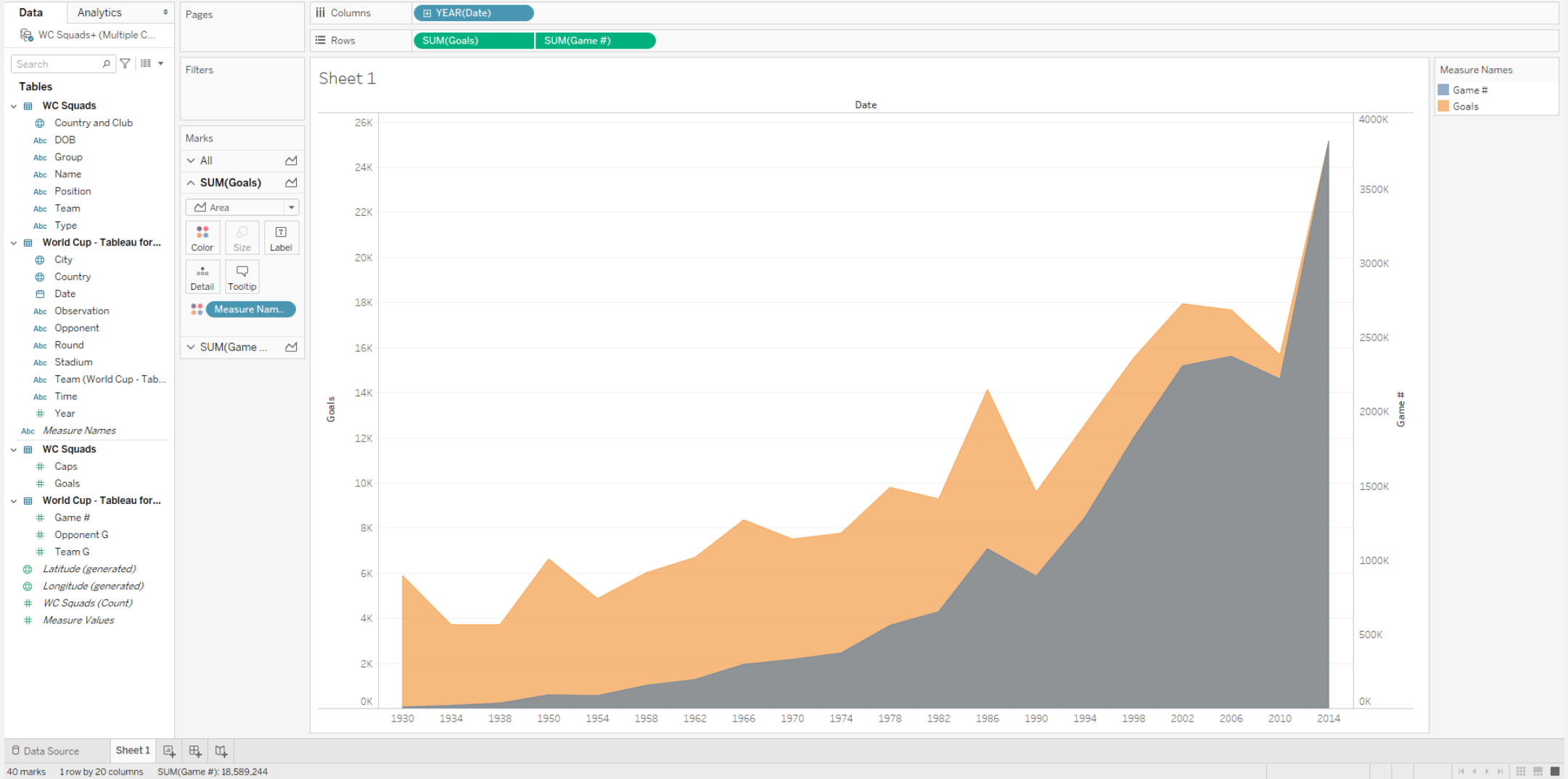


Geographical Graph

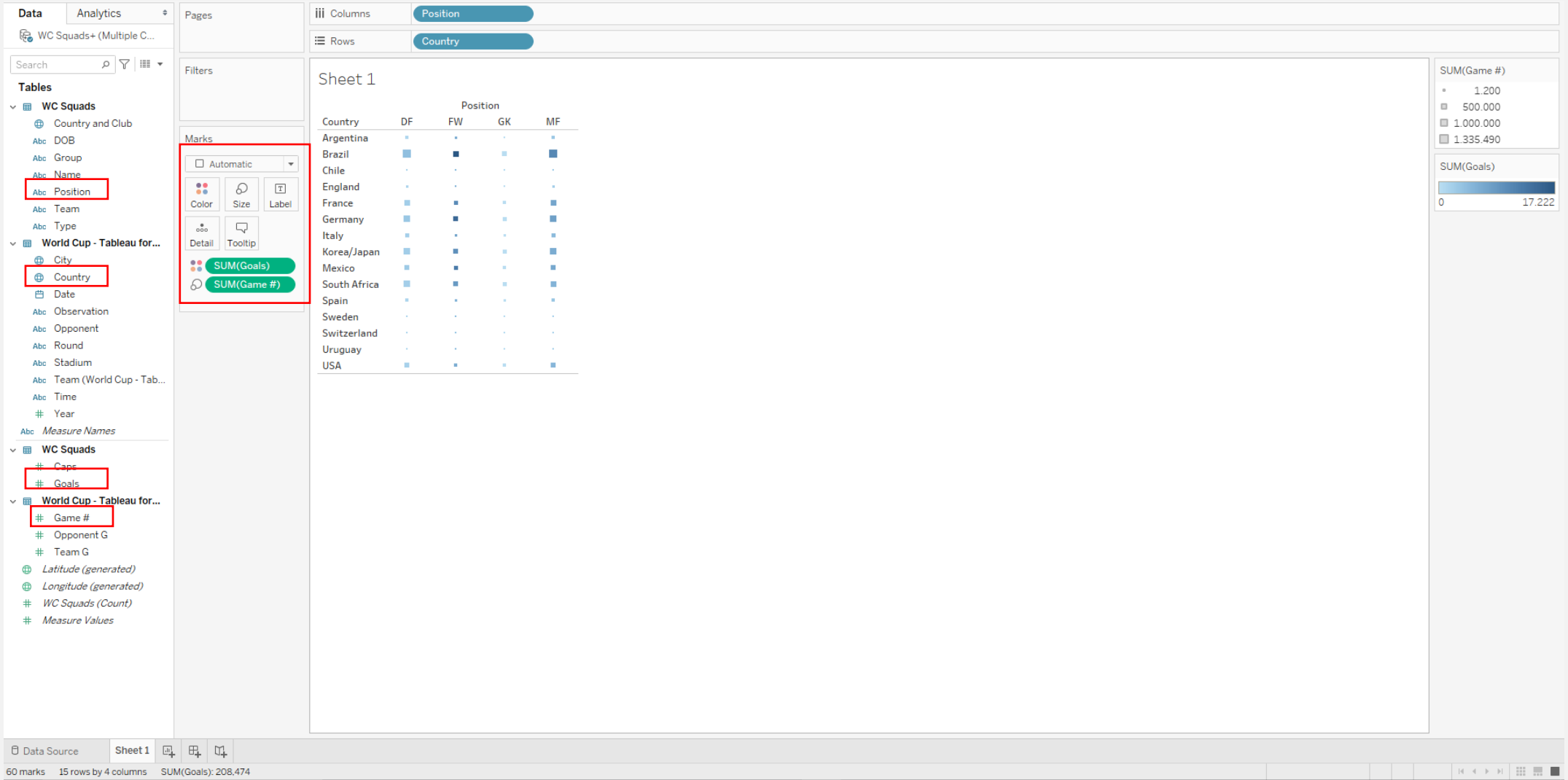
Put the dimension and measurement in the marks card to mark all the point in the map.



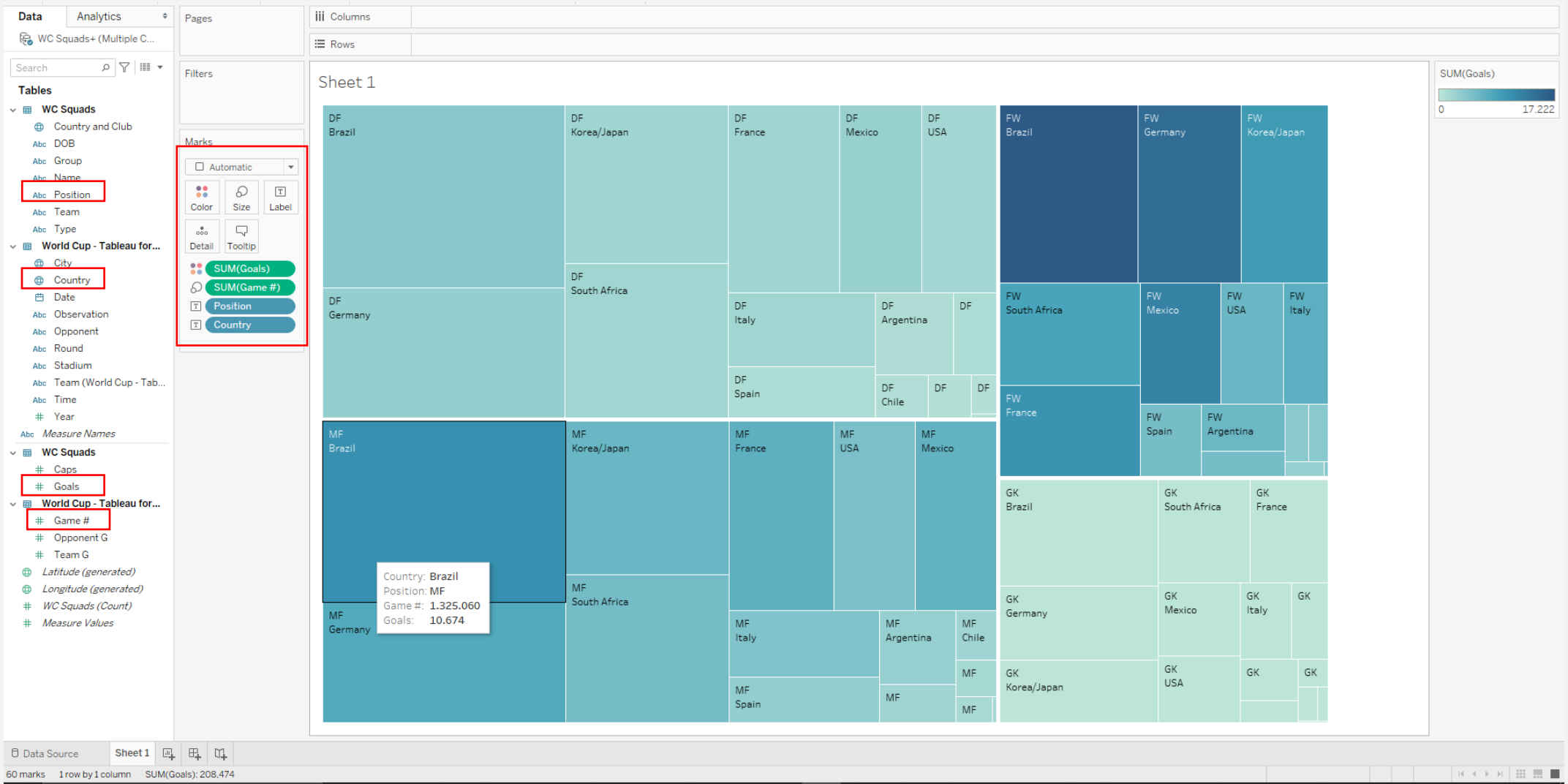
Area Graph – Dual Axis



Heatmap


Data
Analytics
WC Squads+ (Multiple C...
Search
Tables
WC Squads
Country and Club
DOB
Group
Name
Position
Team
Type
World Cup - Tableau for...
City
Country
Date
Observation
Opponent
Round
Stadium
Team (World Cup - Tab...
Time
Year
Measure Names
WC Squads
Caps
Goals
World Cup - Tableau for...
Game #
Opponent G
Team G
Latitude (generated)
Longitude (generated)
WC Squads (Count)
Measure Values
Pages
Filters
Marks
Automatic
Color
Size
Label
Detail
Tooltip
SUM(Goals)
SUM(Game #)
Columns
Position
Rows
Country
Sheet 1
Position
Country
DF
FW
GK
MF
Argentina
Brazil
Chile
England
France
Germany
Italy
Korea/Japan
Mexico
South Africa
Spain
Sweden
Switzerland
Uruguay
USA
SUM(Game #)
1.200
500.000
1.000.000
1.335.490
SUM(Goals)
0
17.222
Data Source
Sheet 1
60 marks
15 rows by 4 columns
SUM(Goals): 208.474

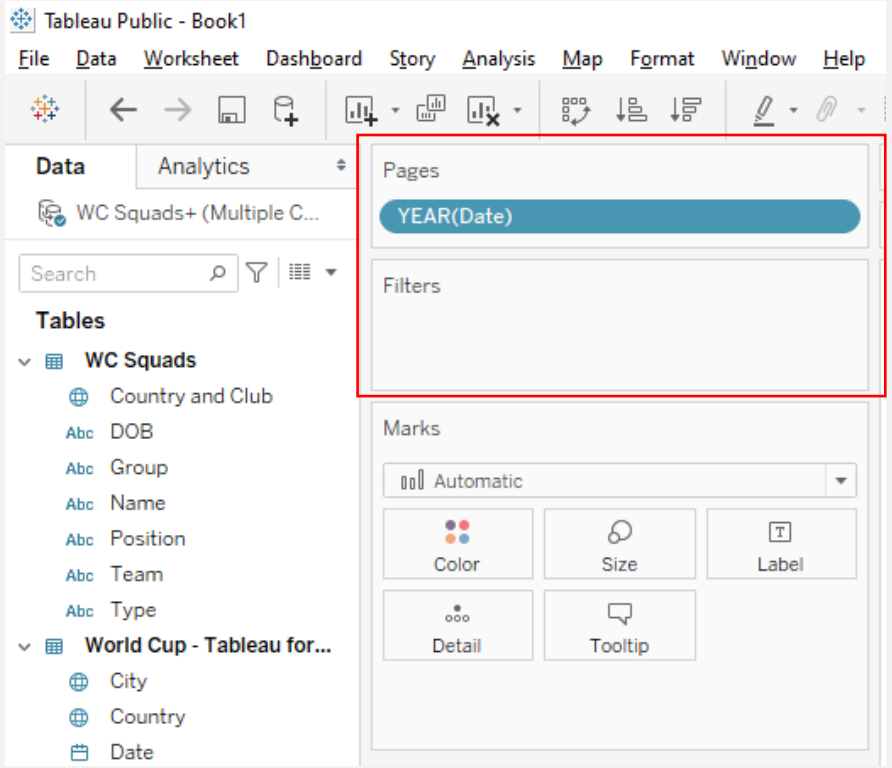
Tree Map


Data
Analytics
WC Squads+ (Multiple C...
Search
Tables
WC Squads
Country and Club
DOB
Group
Name
Position
Team
Type
World Cup - Tableau for...
City
Country
Date
Observation
Opponent
Round
Stadium
Team (World Cup - Tab...
Time
Year
Measure Names
WC Squads
Caps
Goals
World Cup - Tableau for...
Game #
Opponent G
Team G
Latitude (generated)
Longitude (generated)
WC Squads (Count)
Measure Values
Pages
Filters
Marks
Automatic
Color
Size
Label
Detail
Tooltip
SUM(Goals)
SUM(Game #)
Position
Country
Columns
Rows
Sheet 1
DF Brazil
DF Korea/Japan
DF France
DF Mexico
DF USA
DF Germany
DF South Africa
DF Italy
DF Argentina
DF
DF Spain
DF Chile
DF
DF
FW Brazil
FW Germany
FW Korea/Japan
FW South Africa
FW Mexico
FW USA
FW Italy
FW France
FW Spain
FW Argentina
MF Brazil
MF Korea/Japan
MF France
MF USA
MF Mexico
MF Germany
MF South Africa
MF Italy
MF Argentina
MF Chile
MF
MF Spain
MF
MF
Country: Brazil
Position: MF
Game #: 1.325.060
Goals: 10.674
GK Brazil
GK South Africa
GK France
GK Germany
GK Mexico
GK Italy
GK
GK Korea/Japan
GK USA
GK
GK
SUM(Goals)
0
17.222
Data Source
Sheet 1
60 marks
1 row by 1 column
SUM(Goals): 208.474

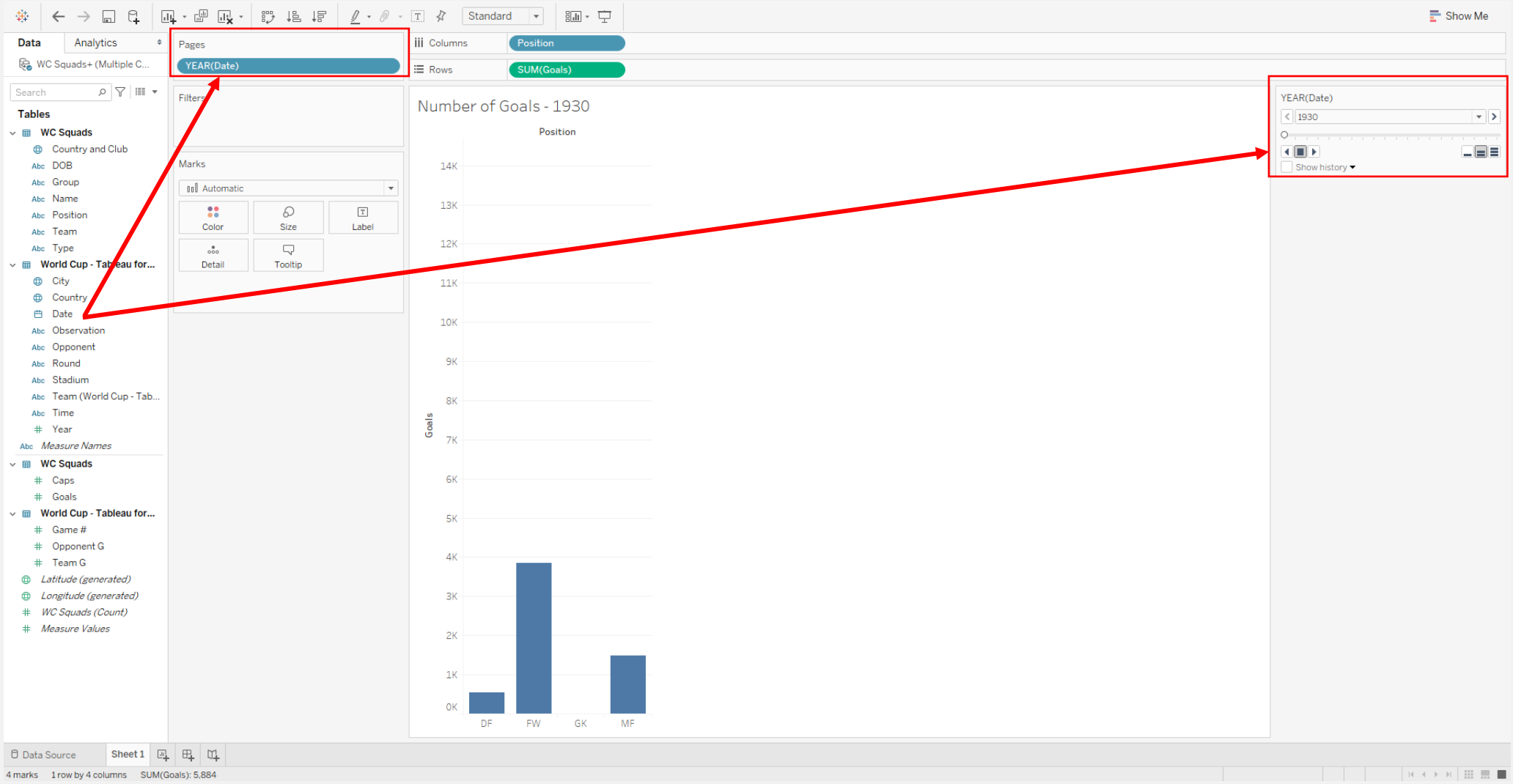
Worksheet – Pages & Filter

- **Pages** : Tableau allows you to analyze your data with the Pages shelf. Sometimes you may want to analyze your data based on the individual values contained within a field. This type of analysis may help you determine whether you're doing better over time.

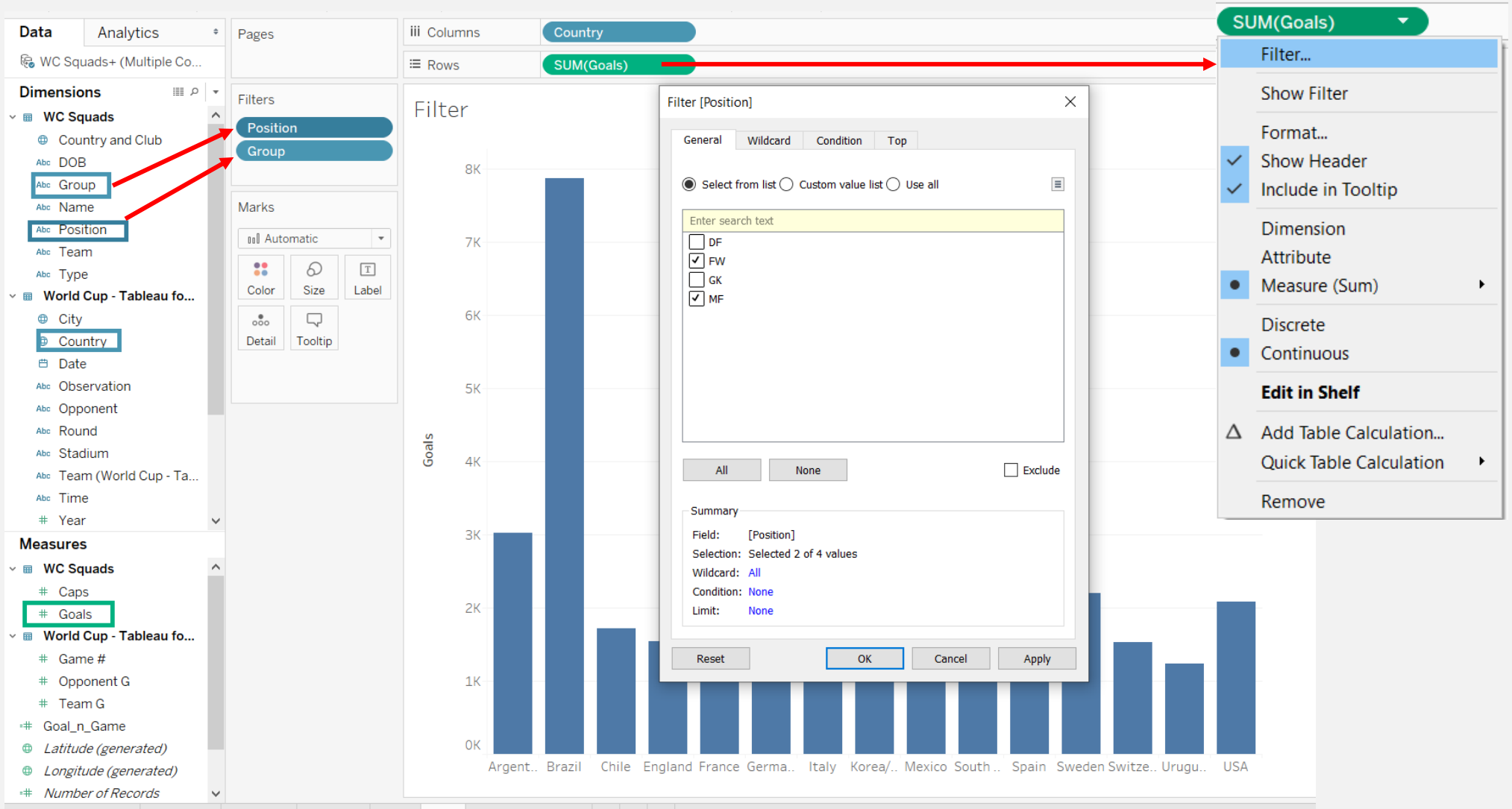
- **Filter** : The filters can be applied in a worksheet to restrict the number of records present in a dataset. There are various way to filtering the data in the worksheets, depend on the user intention



Work Sheet Pages



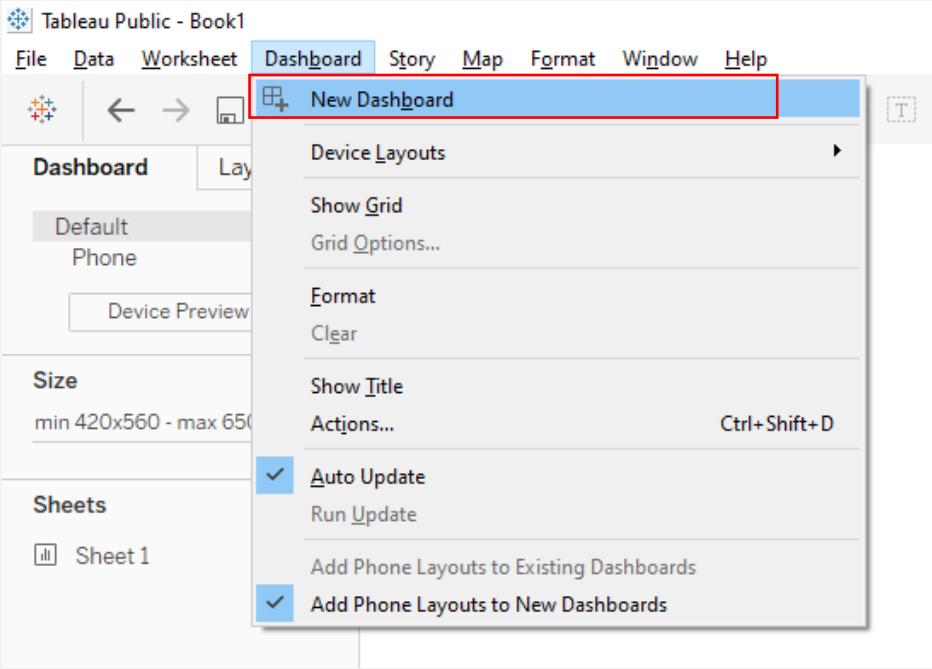
Work Sheet Pages



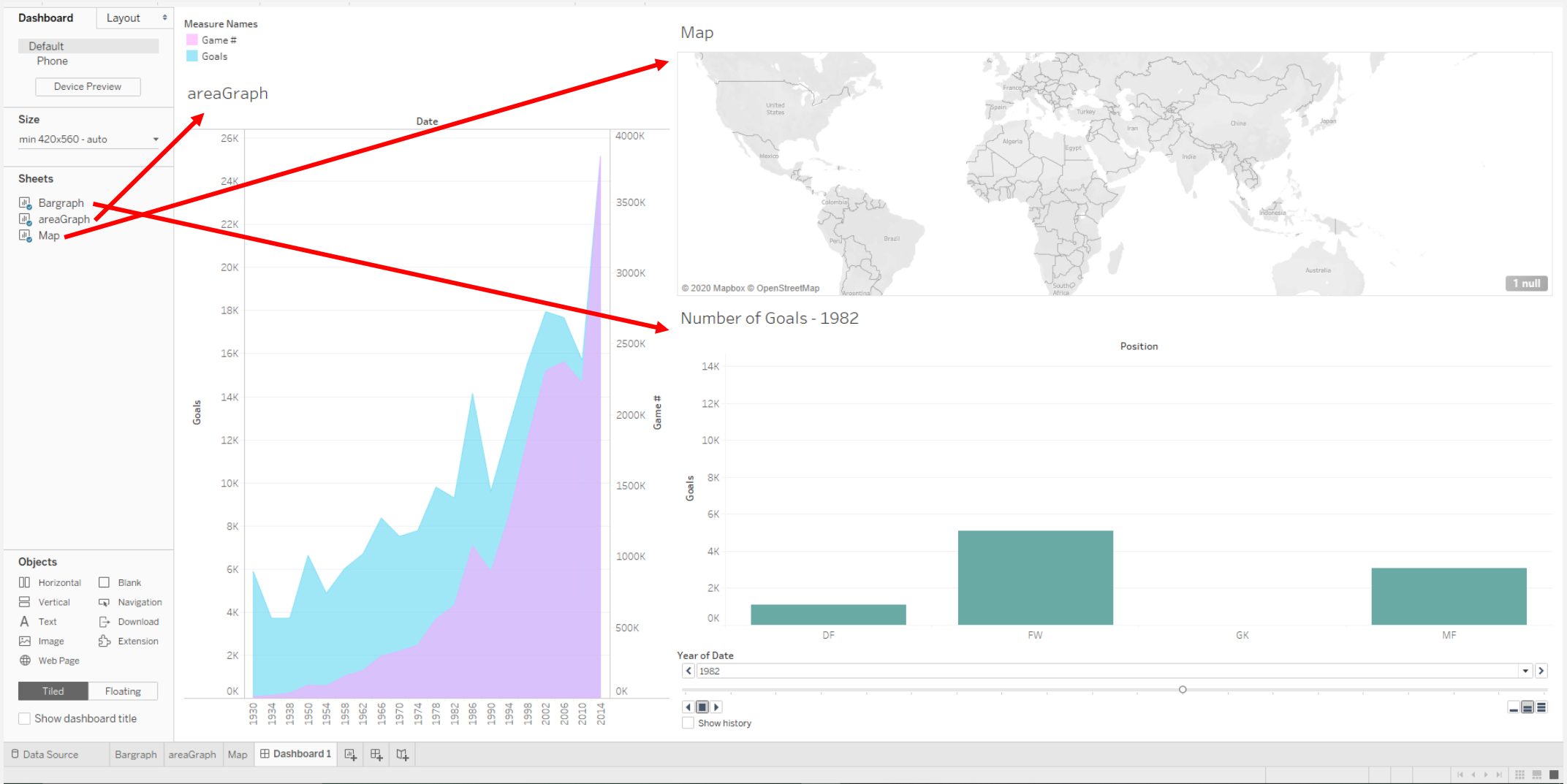
Dashboard

- A dashboard is a consolidated display of many worksheets and related information in a single place. It is used to compare and monitor a variety of data simultaneously.

- Each view you add to the dashboard is connected to its corresponding worksheet. So when you modify the worksheet, the dashboard is updated and when you modify the view in the dashboard, the worksheet is updated.



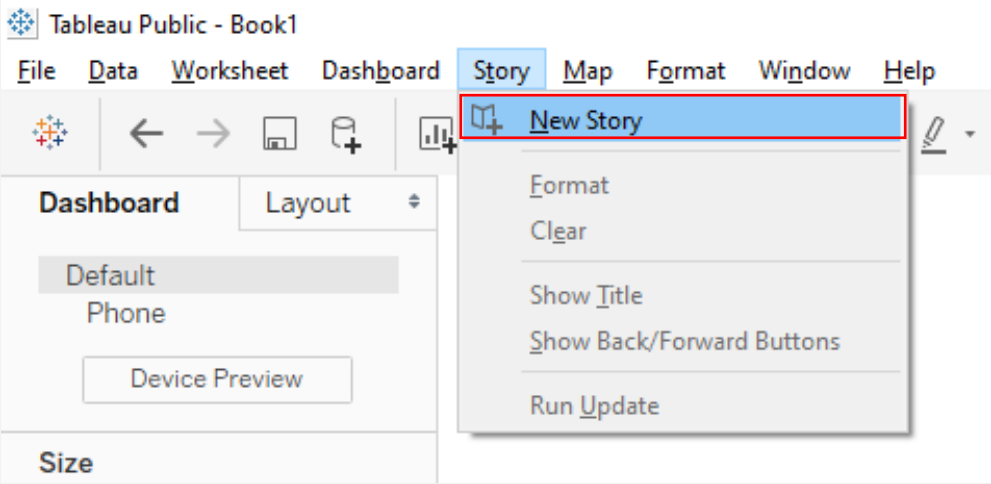
Dashboard


Dashboard
Layout
Default
Phone
Device Preview
Size
min 420x560 - auto
Sheets
Bargraph
areaGraph
Map
Objects
Horizontal
Blank
Vertical
Navigation
Text
Download
Image
Extension
Web Page
Tiled
Floating
Show dashboard title
Measure Names
Game #
Goals
areaGraph
Date
Goals
Game #
Map
© 2020 Mapbox © OpenStreetMap
1 null
Number of Goals - 1982
Position
Goals
DF
FW
GK
MF
Year of Date
1982
Show history
Data Source
Bargraph
areaGraph
Map
Dashboard 1

Story

- In Tableau, a story is a sequence of visualizations that work together to convey information. A Story is a sheet, so the methods to create, name, and manage worksheets and dashboards also apply to stories.

- At the same time, a story is also a collection of sheets, arranged in a sequence. Each individual sheet in a story is called a story point.

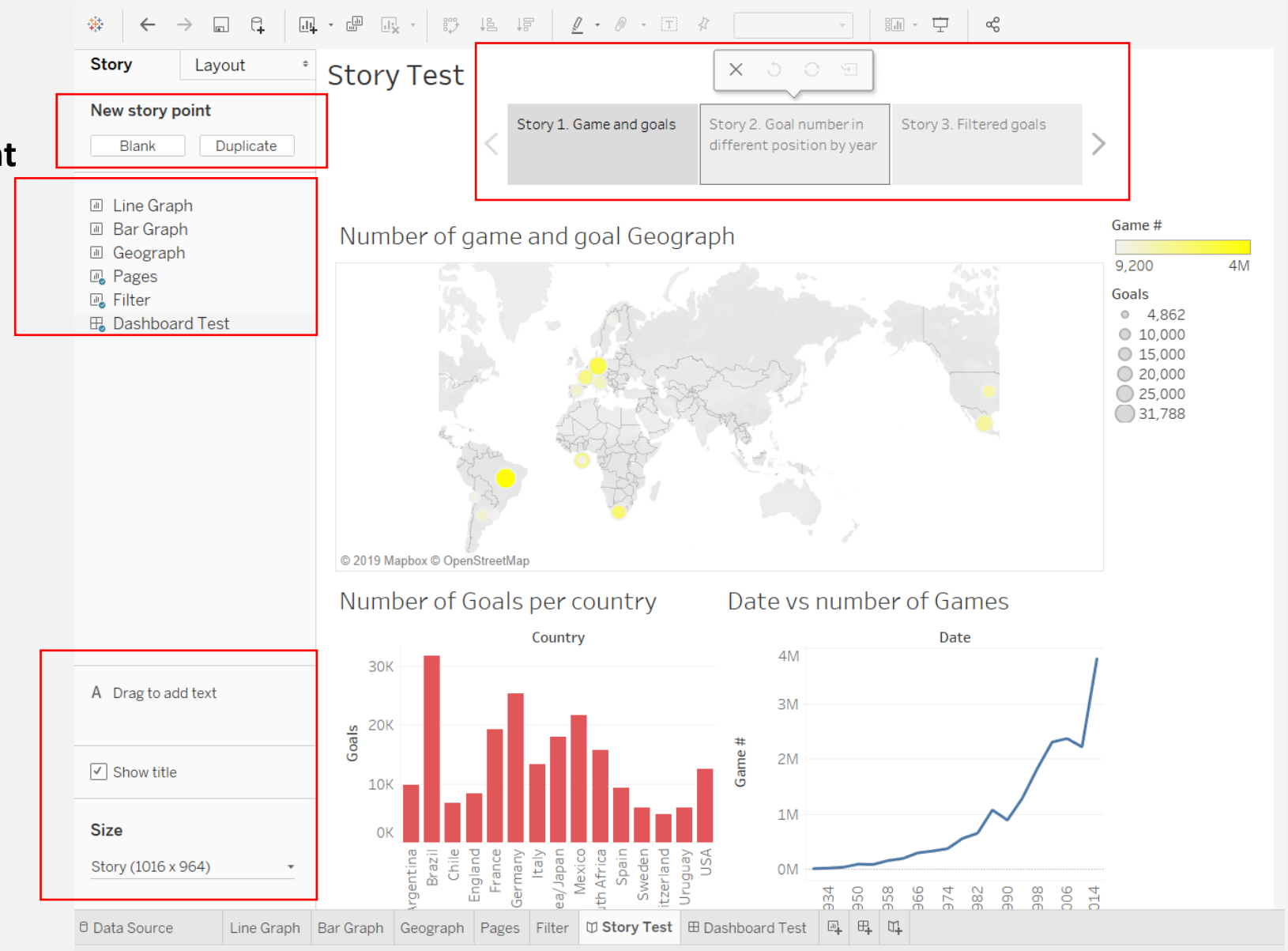


Story

Use this to add more story point

All sheet and dashboard created would be listed here. Just drag and drop to create the story

Use this option to tweak the story size or adding text

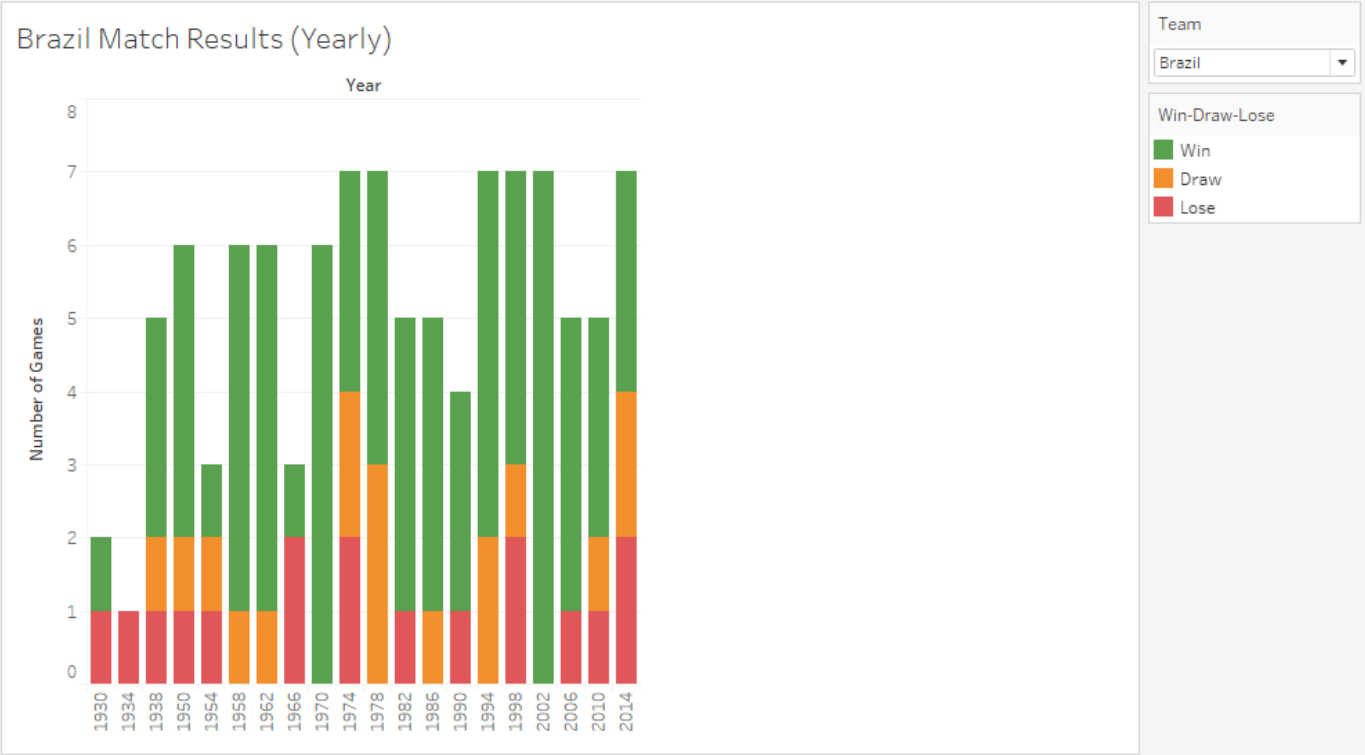


Exercise

Google Search Console
Couverture > Introuvable (404)
Vue d'ensemble
Inspection de l'URL
Performances
Résultats de recherche
Discover
Index
Couverture
Sitemaps
Suppressions
Améliorations
Signaux Web essentiels
Ergonomie mobile
Logos
Extraits d'avis
Toutes les pages connues
Introuvable (404)
État : Exclues
EN SAVOIR PLUS
Pages concernées
52
Exemples
URL
MacBook Air

Exercise 1

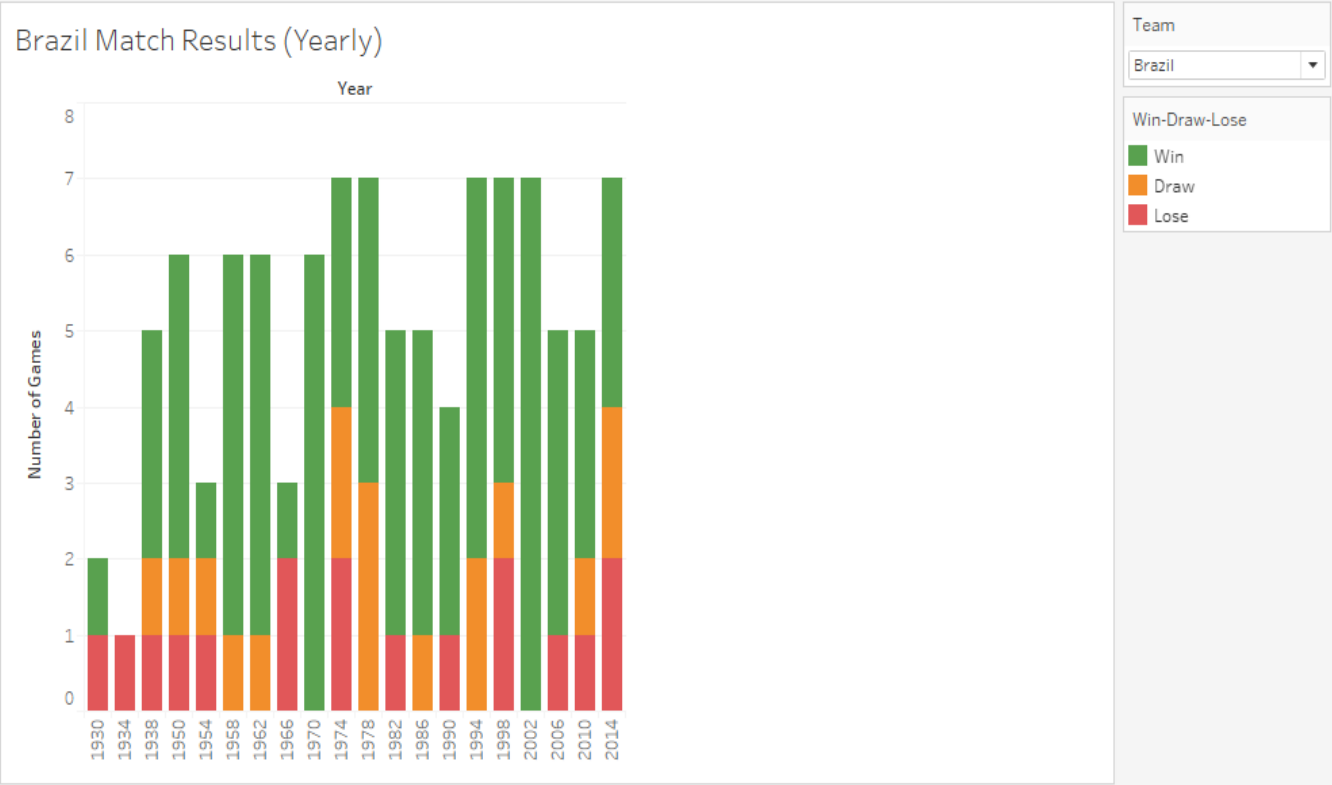
- Buatlah sebuah ***stacked bar plot*** yang menampilkan **jumlah kemenangan**, **kekalahan, dan hasil seri** yang diperoleh oleh suatu tim peserta Piala Dunia di tiap tahunnya.



Exercise 1 (cont'd)

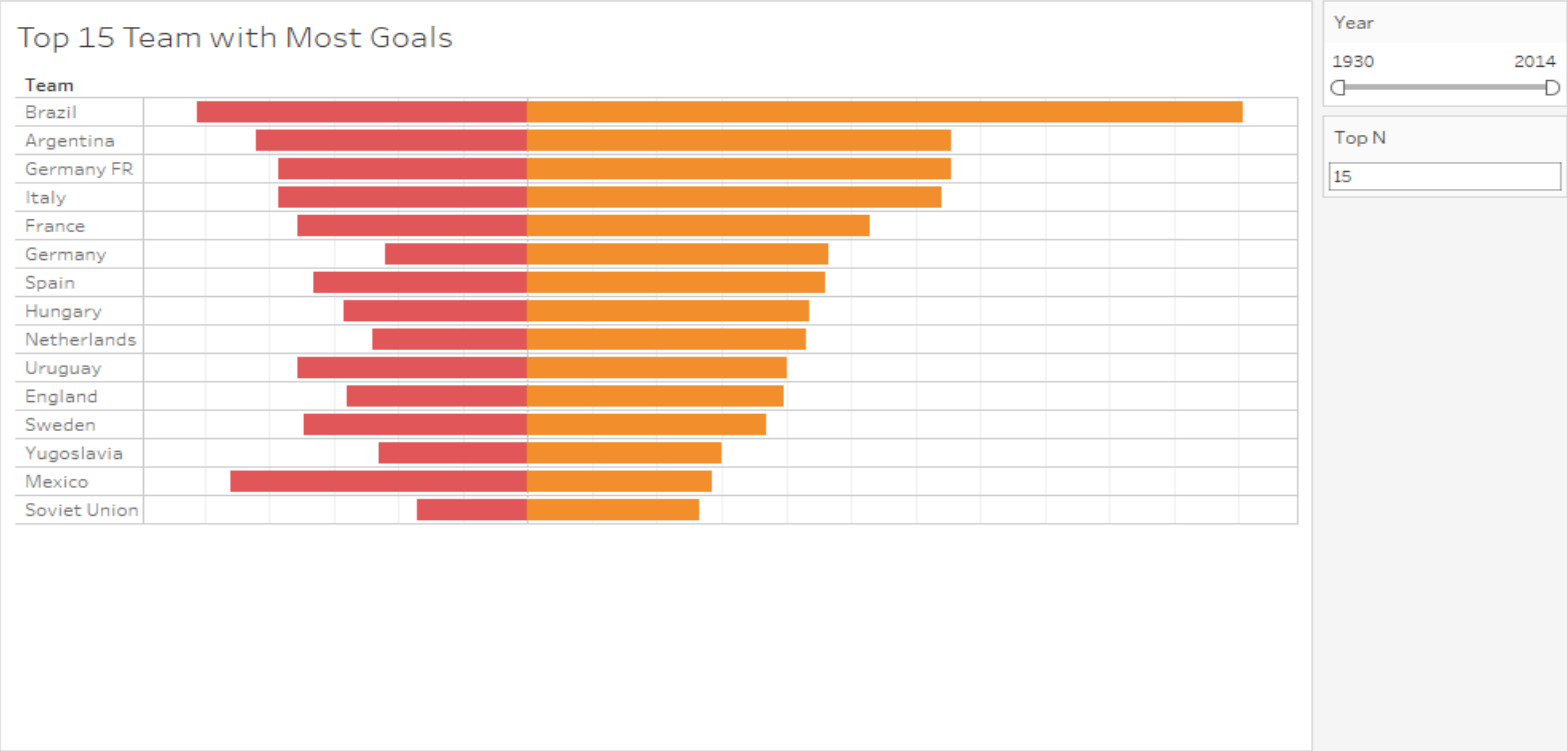
Ketentuan:

1) Tiap *stacked bar* menampilkan **jumlah kemenangan, kekalahan, dan hasil seri** yang diperoleh oleh suatu tim dalam satu tahun.
2) **Tim yang ditampilkan** di dalam *chart* dipilih berdasarkan ***filter*** (*chart* hanya menampilkan data dari satu tim)
3) Terdapat ***color legend*** untuk menjelaskan makna dari tiap warna
4) **Judul *chart*** menampilkan **nama tim** yang sedang ditampilkan datanya di dalam *chart*



Exercise 2

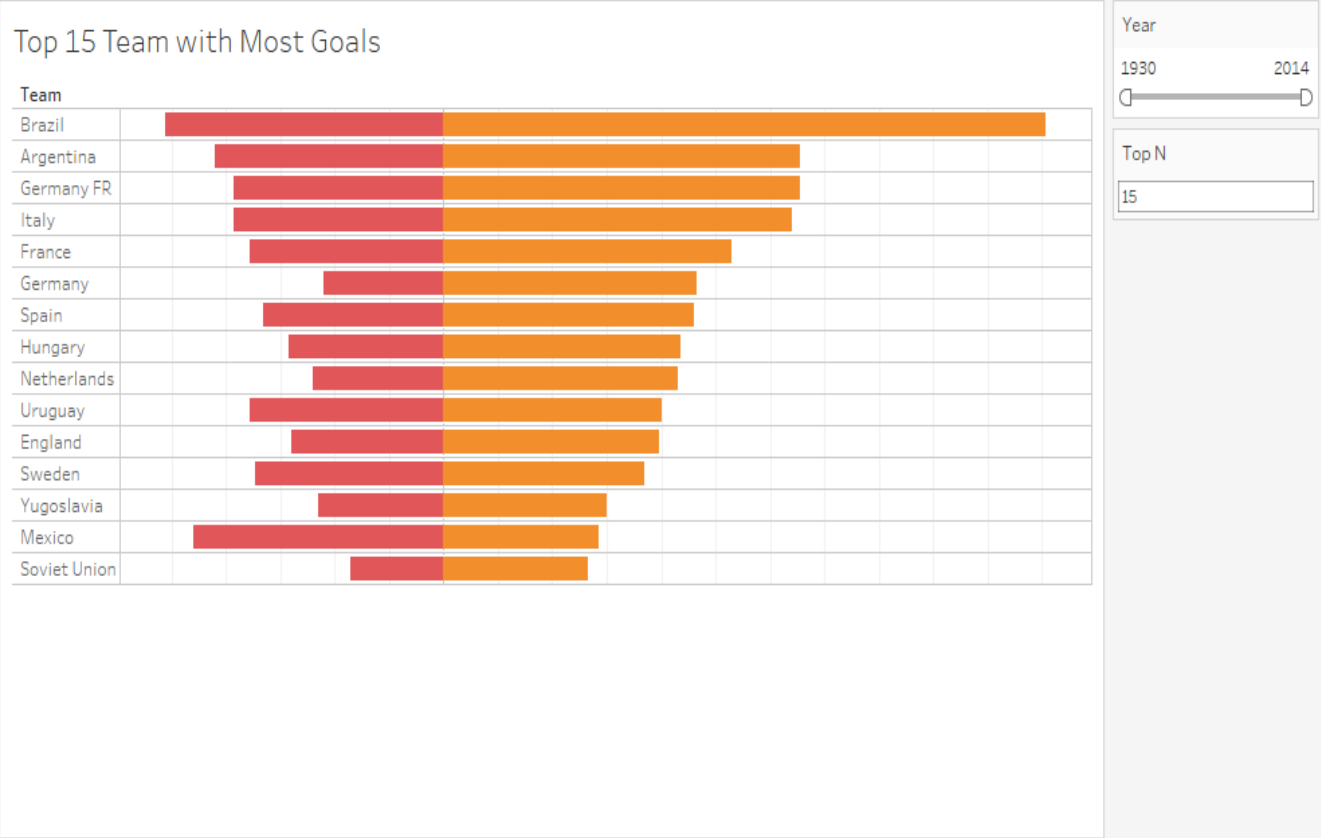
Buatlah sebuah ***horizontal bar plot*** yang menampilkan **jumlah gol yang dicetak** oleh setiap tim dan **jumlah gol yang masuk ke gawang** tim tersebut.



Exercise 2 (cont'd)

Ketentuan:

1) *Bar plot* untuk menampilkan **jumlah gol yang dicetak dan jumlah gol yang masuk ke gawang suatu tim** diletakkan **bersebelahan**
2) **Nama tim** diurutkan menurun (***descending***) berdasarkan **jumlah gol yang dicetak**
3) *Chart* hanya **menampilkan *n* tim dengan jumlah gol terbanyak**, di mana *n* dapat diatur menggunakan ***filter Top N***
4) Jumlah gol yang ditampilkan dapat diatur menggunakan ***filter* tahun** pelaksanaan Piala Dunia (***Year***)
5) **Judul *chart*** secara otomatis berubah sesuai dengan **jumlah tim yang sedang ditampilkan** di dalam *chart.*



Exercise 3

- Cobalah membuat Dashboard dan Story pada Tableau menggunakan Tesla Sales Data yang diberikan

Thank You!

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